DELIVERABLE D1 · INCEPTION, REQUIREMENTS & ENVIRONMENT SETUP

Inception Report

Lesotho Child Helpline 116 — OpenCHS and Primero CPIMS+ Integration

PREPARED FOR	UNICEF Lesotho, MGYSD / Ministry of Social Development, Child Helpline 116 stakeholders, Primero technical team
PREPARED BY	BITZ IT Consulting Ltd — OpenCHS project team
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Deliverable	D1 — Inception, Requirements & Environment Setup (Workstream 1; contract item 10)
Prepared for	UNICEF Lesotho, MGYSD / Ministry of Social Development, Child Helpline 116 stakeholders, Primero technical team
Prepared by	BITZ IT Consulting Ltd, OpenCHS project team
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Reading this report

The main report presents the inception findings, the proposed approach, responsibilities and the decisions required. The annexes contain the detailed requirements, specifications and supporting evidence for specialist reviewers.

OpenCHS is the Child Helpline's case-recording system. Primero CPIMS+ is the child-protection case-management system used by the receiving Ministry teams. **ABR ID** means the birth-registration identifier discussed during inception. **SOPs** are standard operating procedures: the written instructions for handling cases. **UAT** means user acceptance testing: testing by staff to confirm that the systems support their work. **API** (application programming interface) means the software interface through which the two systems exchange information. **D1–D4** identify the four contracted deliverables. Other codes, such as A-014 or R-011, let readers find the related action or risk; they are not technical instructions. Technical terms are used in their standard sense: authentication (how one system proves it is authorised to connect to another), retry (an automatic further attempt to deliver a referral), regression testing (testing that existing functions still work after a change), webhook (an automatic notification sent from one system to another) and go-live (the start of live operational use). Technical terms are used in their standard sense: authentication (how one system proves it is authorised to connect to another), retry (an automatic further attempt to deliver a referral), regression testing (testing that existing functions still work after a change), webhook (an automatic notification sent from one system to another) and go-live (the start of live operational use). Technical terms are used in their standard sense: authentication (how one system proves it is authorised to connect to another), retry (an automatic further attempt to deliver a referral), regression testing (testing that existing functions still work after a change), webhook (an automatic notification sent from one system to another) and go-live (the start of live operational use). Technical terms are used in their standard sense: authentication (how one system proves it is authorised to connect to another), retry (an automatic further attempt to deliver a referral), regression testing (testing that existing functions still work after a change), webhook (an automatic notification sent from one system to another) and go-live (the start of live operational use). Technical terms are used in their standard sense: authentication (how one system proves it is authorised to connect to another), retry (an automatic further attempt to deliver a referral), regression testing (testing that existing functions still work after a change), webhook (an automatic notification sent from one system to another) and go-live (the start of live operational use). Technical

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"Confirmed" means a decision was recorded in a meeting. "Proposed" means it still needs agreement. "Pending" means more information or a decision is needed. None of these terms means this report has been formally approved.

Status classifications used in this report

Classification	Meaning in this report	Equivalent in the completion inventory
Confirmed	Agreed and recorded in session minutes (principally Session 3, 26 October 2026). A confirmed business decision is not formal approval of this report or a contract variation.	Resolved
Approved	Formally approved in writing by a named, authorised approver. No item in this report is currently Approved.	—
Completed	An action or output for which completion evidence has been received.	—
Proposed	A BITZ technical, planning or governance proposal awaiting review.	Proposed
Pending Confirmation	Identified requirement or decision awaiting confirmation from a named party.	Pending
Dependency	An external input that must be received before dependent work can proceed or be accepted.	Pending

Items intentionally outside the baseline are described as **Excluded (Confirmed)** where a session decision supports the exclusion, or **Proposed exclusion** where the boundary is a BITZ scoping proposal (inventory status *Deferred*).

Version history

Version	Date	Change
0.4	26 Oct 2026	Sessions 1–2 and the Primero technical response [TR1]; archived
0.5	26 Oct 2026	Session 3 decisions incorporated; archived
0.6	27 Oct 2026	Contract context, requirement evolution, Primero v2.16.0 API findings, risks R-009–R-012
0.7	27 Oct 2026	Full restructure into a formal inception report: executive summary, methodology, categorised requirements, architecture diagrams, implementation methodology, work plan, governance and RACI, scope-change section, security framework, UAT plan, D1 completion matrix and annexes. No Session 3 decision changed. See the Improvement Log.
0.8	28 Oct 2026	Reorganised into the agreed 12-section order; detailed requirement records moved to Annex J; evidence corrections retained; references updated. No business decision or approval status changed.
0.9	28 Oct 2026	Plain-language main report for programme and Ministry readers; exact twelve-section order retained; detailed design and safeguards preserved in Annex K.
1.0	28 Oct 2026	Aligned planning with owner-supplied TICAD-2026-001: detailed monthly timetable, January user-testing approval, February deployment and handover, reporting and support obligations; contract discrepancies retained for confirmation.
2.0	28 Oct 2026	Re-issued under the BITZ IT Consulting Ltd corporate identity: branded cover page, BITZ Blue headings and table styling, logo header and page-numbered

Version	Date	Change
		footer, unified Arial typography and an automatic table of contents. No content, decision or approval status changed from version 1.0.
2.1	28 Oct 2026	Editorial review for a formal inception-report register: standard technical terminology restored (authentication, retry, regression testing, configuration, go-live, webhook, data mapping); internal workspace and evidence-filing language replaced; attendance reconciliation detail condensed; section 5 to 10 headings formalised; citations standardised to the source register; duplicate diagram labels removed; figures renumbered in order of appearance; section 5 expanded with the agreed referral lifecycle, business rules and pending matters drawn from Annexes A and D. No decision, requirement, risk or approval status changed.

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1 Executive summary

1.1 Purpose of this report

This inception report constitutes Deliverable D1 (Inception, Requirements and Environment Setup) under contract TICAD-2026-001 [CT1]. It presents the findings of the inception phase, the agreed requirements baseline, the proposed technical approach, the work plan and the governance arrangements for integrating OpenCHS with Primero CPIMS+, and it identifies the decisions required from UNICEF and the Ministry. The delivery timetable and obligations follow the TICAD-2026-001 contract text. The earlier Technical Scope Brief cites contract reference 43427866 Amendment 3 [TB]; UNICEF and BITZ are requested to confirm the relationship between the two references and to provide the signed governing document.

The minutes of the inception sessions remain the primary record of the discussions and agreements. Rather than restate them, this report cites each adopted position to the relevant minute section or decision reference. Bracketed codes refer to the source register in Annex I.

1.2 Basis of the report

Record	Ref.	Relevance to this report
Technical Scope / TOR Brief, prepared for the inception meeting of 24 September 2026	[TB]	Contractual scope, workstreams and inception questions
Session 1 Minutes — Business and Workflow, 8 October 2026	[M1]	Helpline workflow and referral data
Session 2 Minutes — Technical Architecture, 9 October 2026	[M2]	CPIMS+ referral mechanism; decisions D-001–D-004
Session 3 Minutes — Closing Requirements, 26 October 2026	[M3]	Decisions D-005–D-018; actions A-013–A-025
Session 3 Action Tracker, 26 October 2026	[AT3]	Action owners, due dates and status
Primero Technical Response to the BITZ inception questionnaire of 2 October 2026	[TR1]	Environment, authentication, error handling and test data
Annotated Requirements Checklist (Session 3)	[RC1]	ABR format, UAT participants and sign-off, data protection reference
Primero API v2 documentation and source, release v2.16.0	[PA]	API operations and capabilities (Annex B)
Contract-Aligned Delivery Plan v0.3	[DP]	Milestone months and approval checkpoints

1.3 Agreed requirements

Session 3 closed most of the requirements carried over from Sessions 1 and 2 [M3, Executive summary]. This report adopts the decisions in the Session 3 decisions log without amendment as its agreed requirements. They are reproduced in Annex E and traced to requirements in Annex J.

Area	Position adopted	Recorded in
Referral initiation	Manual, by the handling agent	[M3 section 3]; D-006
Routing	Agent selects a CPIMS+ service; Ministry distributes	[M3 section 13]; D-008
Integration direction	Two-way for referred cases	[M3 section 11]; D-007

Area	Position adopted	Recorded in
Reporter and child identification	Anonymous reporter permitted; child identifying details required	[M3 section 6]; D-010, D-011
Referral information set	Field set as presented, including attachments; narrative required at referral; case plan excluded	[M3 section 5]; D-012, D-013
ABR ID	Optional; the supplied NICR number format will be used for the first checks	[M3 sections 5, 9]; D-005, D-014; [RC1] item 6
System connection	A direct software connection or OpenFN; X-Road only for the separate Home Affairs checks	[M3 section 9]; D-015
Acknowledgement, API and UAT	Distinguish system receipt from a person accepting the referral; use the supplied connection specification; Ministry leads user testing	[M3 section 10]; D-016, D-017, D-009
Test environment	Primero global demonstration system if the Lesotho test system is not ready	[M3 section 8]; D-018

Session decisions are agreements recorded in the minutes. They are not formal approval of this report, and they are not a contract variation.

1.4 Matters remaining open

Matter	Position	Reference
Referrals that cannot reach CPIMS+ during service interruptions	Left open in Session 3 for follow-up or testing	[M3 section 7]; A-021
Lesotho test system readiness and access	Checklist incomplete; access due 28 October	[M3 section 8]; A-015, A-016
National referral pathways and SOPs	SOP received; current applicability pending Ministry confirmation. Separate national pathways still outstanding	[M3 section 4]; A-013, A-014
Data protection basis	Material to be shared by UNICEF; desk research by BITZ	[M3 section 12]; A-017, A-022
CPIMS+ service list	To be confirmed	[M3 section 13]; A-018
Integration authentication	The method for the systems to prove they are authorised to connect is not yet agreed	[TR1 section 3]; API-002
Return-update mechanism and status vocabulary	Direction agreed; mechanism and status values not defined	D-007; [M2 section 5]; RT-006, RT-007
Minimum information for emergency referrals	Agreed in principle; content not defined	[M2] D-003; WF-004
Consent at referral	Not discussed in the sessions; required by the standard Primero referral check	[PA]; Annex B.6; R-011

The contractual effect of two-way updates, confirmation of the governing contract and the treatment of missed milestones need decisions (section 2.8). The supplied TICAD text explicitly states July 2026–February 2027; the uncertainty is its relationship to the earlier contract reference and signed status.

1.5 Proposed implementation approach

BITZ proposes a secure, authenticated connection between OpenCHS and Primero that queues undelivered referrals for retry and logs each exchange (section 6). The proposed design uses a stable reference for each case to prevent duplicates when delivery is attempted again. This safeguard must be confirmed and tested for

Lesotho [PA]. System receipt, acceptance and case progress are tracked as separate states, in line with D-016. The design is **subject to review and testing by the OpenCHS and Primero technical teams**.

1.6 Current readiness position

Requirements are substantially defined. Implementation readiness depends on external inputs that have not yet been received:

- No Primero test environment access, API credentials or definitions of the information required by the Lesotho system have been received [AT3: A-016].
- As at 28 October 2026, actions A-013, A-017, A-018 and A-020 remain outstanding. Under A-014, the September 2019 SOP has been received; its current applicability is subject to Ministry confirmation [SOP1].
- No stakeholder has formally approved this report (section 12.1).

1.7 Principal risks and dependencies

1. **Environment and access delay** (R-001) — blocks data mapping and connection testing [M3 section 8].
2. **Contract scope change** (R-010) — the contract describes software that sends information from OpenCHS to CPIMS+ [TB section 5], while Session 3 confirmed two-way updates [M3 section 11].
3. **Milestone slippage** (R-009) — the D1 target month (August 2026) has passed, and the D2 window closes on 31 October [DP section 1].
4. **Consent and lawful basis** (R-007, R-011) — the applicable legal instrument is unconfirmed [M3 section 12; RC1 item 11].
5. **Service routing configuration** (R-003) — each CPIMS+ service needs a configured receiving user [PA; Annex B.2].

1.8 Decisions required from UNICEF and the Ministry

#	Decision	Required from
1	Whether two-way updates and attachments are a contract variation, and how effort, milestones and acceptance evidence are adjusted (sections 2.8 and 4.2)	UNICEF (contracting party), with the Ministry
2	Which dates govern milestones and payment, and approval of a revised delivery schedule (section 8)	UNICEF / project governance
3	Named approvers for requirements, technical design, data protection and UAT sign-off (section 9)	UNICEF and Ministry
4	Consent or other lawful basis for sharing, and who may override consent in CPIMS+ (section 7)	Ministry, with UNICEF advice
5	Minimum child-identification dataset and emergency exceptions (WF-004, DATA-005)	Ministry / Child Helpline
6	Direct software connection or OpenFN (D-015 leaves the choice to the Ministry)	Ministry IT, with Primero

The decisions required before design and before integration testing are listed in sections 11.4 and 11.5.

The supplied contract places testing of both systems and UNICEF user-testing sign-off in January 2027, with deployment, training and handover in February. Section 8 separates that original timetable from the proposed

recovery plan. UNICEF review requires five working days; free maintenance continues for 90 days after launch, with arrangements beyond February to be confirmed [CT1].

1.9 Immediate next steps

The actions below follow the Session 3 action register [M3, Action items; AT3]: circulate this draft by 28 October (A-025); follow up outstanding country documents (A-013, A-017, A-020) and confirm received SOP applicability (A-014); obtain environment access and the service list (A-015, A-016, A-018); hold the technical working session on service interruption handling and authentication (A-021, A-023); and table the scope and schedule decisions for governance (section 11).

Overall readiness: this report is ready for stakeholder review; D1 is not yet ready for formal acceptance (section 10.7).

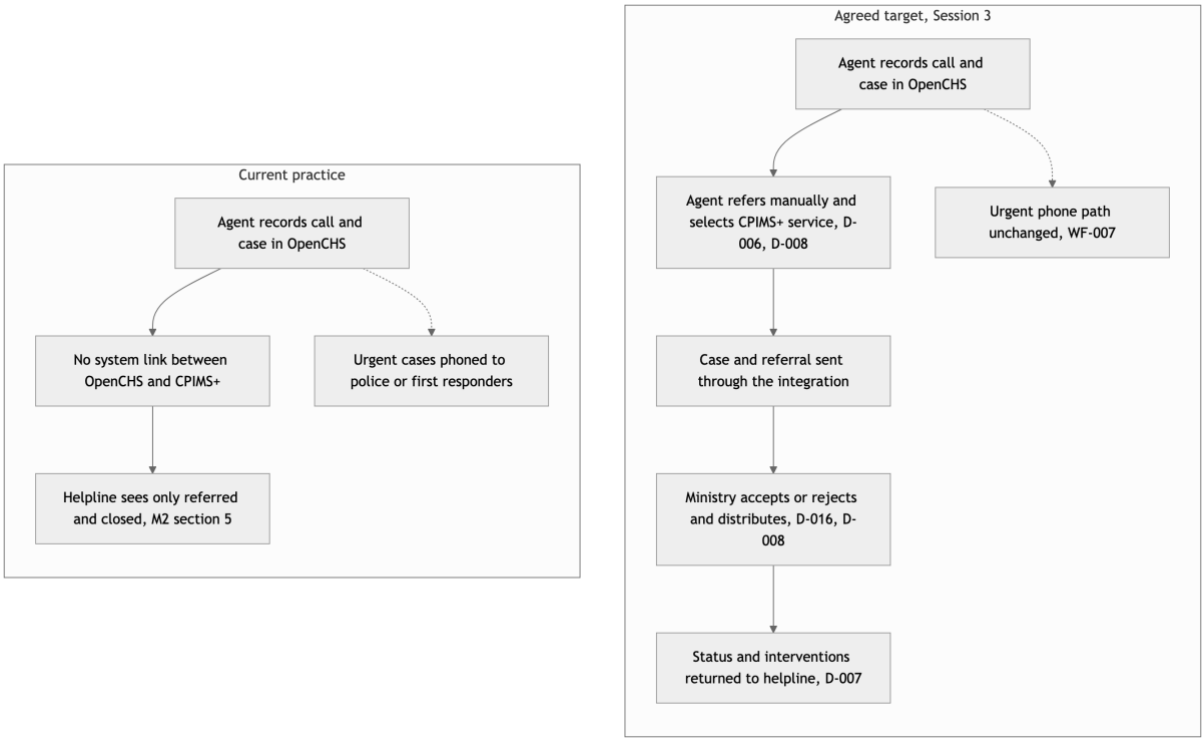
2 Background and contract context

2.1 Operational context

Child Helpline 116 operates a single call centre in Lesotho; any satellite site is a VPN-connected extension of it [M2 section 6]. Helpline agents receive calls, create a contact and case in OpenCHS, and decide whether a case needs referral. Child-protection case management is the Ministry's responsibility and is carried out in Primero CPIMS+. Today, referral between the two depends on manual channels, and the helpline sees only that a case was "referred" and, eventually, "closed" [M2 section 5].

For urgent cases (for example active abuse or an abandoned child), agents phone police or first responders directly. That practice continues unchanged; the integration records and tracks referrals but does not replace urgent calls [M1 section 8; WF-007].

Figure 1 — Current practice and agreed target referral flow. Sources: [M2 section 5] (current status visibility), [M3 sections 3, 11, 13] (target). The target flow is agreed. The way it is implemented is Proposed (section 6).



Aspect	Current practice	Agreed target	Reference
Referral channel	No system link between OpenCHS and CPIMS+	Integration through the Primero API	D-015, D-017
Who initiates	Handling agent	Handling agent, manually	D-006
Destination	—	Selected CPIMS+ service; Ministry distributes	D-008
Acknowledgement	—	System receipt, then human accept or reject	D-016
Status visibility at the helpline	Referred and closed only	Status and interventions returned	[M2 section 5]; D-007
Urgent cases	Direct phone call	Unchanged, plus system tracking	[M1 section 8]; WF-007

2.2 Roles of the two systems

System	Owner / operator	Role in the integration
OpenCHS	Child Helpline 116; maintained by BITZ under a continuing support TOR	Records calls, contacts and helpline cases; captures referral information including the optional ABR ID; initiates referrals; displays referral status and returned updates to helpline staff
Primero CPIMS+	Ministry; Primero technical coordination by Isaiah Amutavi Osotsi	Receives referred cases; Ministry users accept or reject, distribute and manage cases; records services and interventions; source of return updates
OpenCHS integration service / connection gateway	BITZ (Marlon Amunga, Peter Rogendo)	Proposed component that authenticates to Primero, sends referrals and follow-ups, handles retries and duplicates, receives or retrieves return updates, and keeps the audit trail

Both systems sit within the same Ministry, so they connect directly rather than through X-Road (D-015). X-Road is relevant only to verifying child identifiers with the Ministry of Home Affairs, which is a separate coordinated workstream (A-019).

2.3 Justification and intended outcomes

The integration is expected to deliver the outcomes below. These are intended outcomes; measuring their achievement will require baseline data, which was not collected during the inception phase.

Area	Intended outcome
Referral coordination	Referrals reach the Ministry through a consistent, service-based channel aligned with the national referral pathways and SOPs
Case information exchange	The agreed information set (child details, narrative, category, reporter and perpetrator details, attachments) travels with the referral, reducing re-collection
Referral status visibility	Helpline staff can see whether a referral was received, accepted or rejected, and how the case progressed, closing the “referred to closed” gap identified in Session 2
Continuity of case management	New information, including a later finding that a call was a prank, follows the referral as an update to the same case [M1 section 7]
Follow-up and accountability	Return of status and interventions lets the helpline judge effectiveness, not only count referrals [M3 section 11]
Reduced duplicate data entry	Ministry users receive referral data electronically rather than re-keying it
Auditability	Each transmission, acknowledgement and update is logged with identifiers, time, actor and outcome

2.4 Overall objective

To strengthen the Lesotho Child Helpline 116 / OpenCHS environment through requirements definition, configuration, API integration with Primero CPIMS+, testing, user acceptance, capacity building and so-live support, within the continuing support-and-maintenance TOR [TB section 1].

2.5 Specific objectives of the inception phase (D1)

1. Document and confirm the end-to-end referral lifecycle, including two-way feedback.
2. Confirm the information set for referral and establish the field-mapping baseline.

3. Define ABR ID ownership, format and checking rules.
4. Establish authentication, environment, connectivity and security requirements.
5. Define reliability, audit, retry, offline and error-handling expectations.
6. Agree UAT leadership, participants, acceptance and sign-off arrangements.
7. Produce an implementation plan, risk register and decision and action registers.

2.6 Expected results by deliverable

Deliverable	Expected result
D1	Approved inception baseline: requirements, technical design, mapping baseline, environment-setup evidence, work plan and registers
D2	OpenCHS ABR ID field with checks, a referral control enforcing referral-time rules, and referral status display
D3	Working, documented integration client with automated authentication, retry, offline queue, duplicate prevention and audit logging (contract scope is outbound; see section 4.2 for two-way)
D4	Technical testing and regression testing, Ministry-led UAT, training, deployment and handover

2.7 Contractual scope

The text of contract TICAD-2026-001 [CT1] covers 1 July 2026 to 28 February 2027. It provides the planning basis below. The signed original and its general terms have not yet been provided. The earlier scope brief cites a different contract, 43427866 Amendment 3; it remains historical context until UNICEF and BITZ confirm the relationship. The timetable below reflects the contractual targets; it does not indicate that earlier milestones have been completed.

Deliverable	Scope in the supplied contract	Milestone month [CT1]	Evidence for acceptance
D1 Inception, Requirements & Environment Setup	Business and technical requirements, architecture, environments, access, integration workflow, implementation plan	August 2026	Approved inception report, technical design, mapping baseline and environment-setup evidence
D2 ABR ID Field, Checks & Referral Checkbox UI	ABR field, validation rules, referral control, status display	October 2026	Demonstrated field, checks, referral control and unit-test evidence
D3 Outbound REST API Client	automated authentication, retries, offline queue, duplicate prevention, audit logging	November 2026	evidence that the connection follows the agreed specification, using a controlled simulation where necessary
D4 Testing, UAT, Training & Go-Live	Technical testing and regression testing, UAT, training, deployment, handover	February 2027	Signed UAT, training and handover evidence, so-live approval

The supplied contract adds 90 days of free maintenance after launch [CT1 section 6]. Its administration beyond February must be confirmed. Earlier brief-based support activities remain subject to confirmation against the governing contract: minor updates, bug fixing, dependency maintenance, quality assurance after changes, training of IT personnel and system stability [TB section 7]. Regression testing must show that existing OpenCHS functionality is not adversely affected.

2.8 Contractual discrepancies identified

Ref	Discrepancy	Sources	Treatment
CD-01	Contract describes software that sends information from OpenCHS to CPIMS+; Session 3 confirmed two-way updates	[TB sections 2, 9]; D-007	Section 9; R-010; governance decision required
CD-02	TICAD-2026-001 specifies July 2026–February 2027; the earlier brief cites 43427866 Amendment 3 and March 2025–July 2027	[CT1] cover; [TB]	Use supplied TICAD dates for planning; UNICEF/BITZ to confirm governing reference, relationship and signed status
CD-03	The delivery plan describes D2 as a referral checkbox with a “save-event trigger” and the D4 end-to-end test as “OpenCHS save → API call”. Session 3 confirmed a manual referral action with narrative required at referral, not at first save	Delivery plan Items 20, 40; D-006, D-013	Interpret “referral checkbox” as the explicit agent referral control; governance to confirm D2/D4 acceptance wording (Pending Confirmation)
CD-04	D3 names POST /api/v2/cases only; Lesotho routing also needs the separate referral operation, follow-up updates and attachments	Delivery plan Item 30; Annex B	Include within D3 design subject to the CD-01 decision
CD-05	Contract requires ABR format, check-digit rules and at least 10 valid examples; Session 3 provides number structure only	[CT1 sections 3,5]; D-014	Obtain official specification under A-007. Optional ABR entry does not waive the required checks; agree any interim rules in writing
CD-06	Brief expects separate inception report, requirements specification, integration design/field-mapping document and action register	[TB section 11]	This report combines requirements and design; the field-mapping document cannot be completed until configured definitions are received (Dependency)
CD-07	Eight-month contract schedule retains a Week 14 reference; Weeks 2–4 have no confirmed calendar anchor	[CT1 sections 3,5,6,9]; [M1 section 10]	Use explicit months for planning; UNICEF to clarify week references; do not introduce a competing 14-week schedule
CD-08	Access required in August/Week 4 but November availability is anticipated for January tests	[CT1 sections 3,5,6,9]	Record original deadline separately from later assumption; UNICEF/Primero to commit a usable-access date
CD-09	Contract names UNICEF for user-testing sign-off; earlier checklist names MGYSD	[CT1 section 3.3]; [RC1]; D-009	Ministry leads testing; UNICEF executes contractual sign-off; name signatory and confirm Ministry endorsement
CD-10	Ninety days of maintenance after a February launch extends beyond the stated contract period	[CT1 section 6]; cover	Confirm support dates, staffing and administration; do not assume an extension of delivery deadlines
CD-11	Section 6 says electronic submission; section 7 requires electronic and hard copy	[CT1 sections 6–7]	Plan both formats and UNICEF templates unless UNICEF confirms otherwise
CD-12	Contract requires faults resolved to UNICEF satisfaction; earlier report proposes waivers	[CT1 section 3.3]; report section 10	Proposed waiver thresholds do not establish permission to accept unresolved faults; obtain UNICEF agreement

3 Inception activities and findings

3.1 Approach

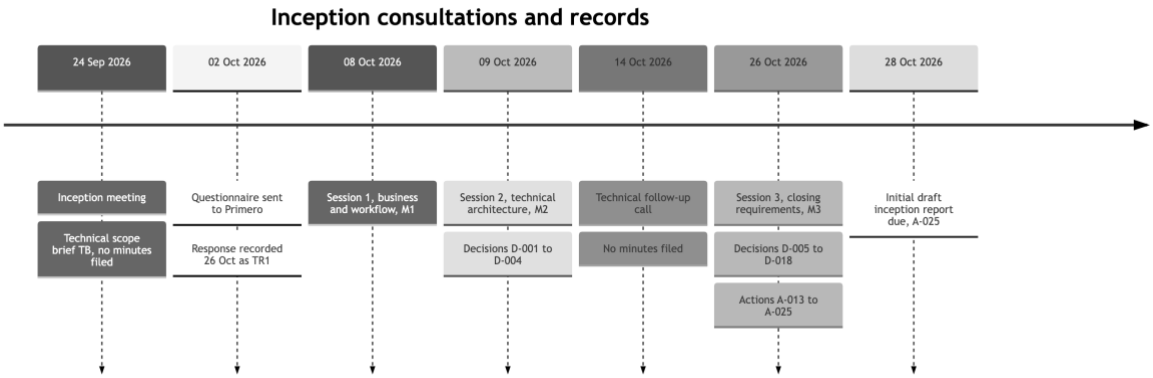
Method	What was done	Evidence
Document review	Technical scope brief; contract-aligned delivery plan; Session 1–3 minutes [M1–M3]; Primero technical response [TR1]; annotated requirements checklist [RC1]; earlier technical reference material	Annex I
Stakeholder consultations	Three facilitated sessions with Child Helpline, Ministry, Primero, UNICEF (Session 3) and BITZ participants	Session minutes (section 3.2)
Business workflow analysis	OpenCHS intake walkthrough (Session 1); referral lifecycle walk-through (Session 2); lifecycle presentation and confirmation [M3 section 3]	[M1 section 1]; [M2 section 3]; D-006
Technical requirements discussion	Environments, routing, duplicates, audit, status visibility (Session 2); acknowledgement, API, service interruption, test system (Session 3)	[M2 sections 4–11]; [M3 sections 7–10]
Review of Primero API capability	Live CPIMS+ referral demonstration [M2 section 4]; Primero technical response [TR1]; review of the standard Primero v2.16.0 API documentation and source code	Annex B
Requirements checks	Session 3 recap and item-by-item confirmation, with decisions logged	D-005 – D-018
Decision consolidation	Decision register continued from Sessions 1–2; superseded positions identified	Annex E
Dependency identification	Action tracker, completion inventory, risk register	Annexes F and G

3.2 Consultations conducted

Date	Event	Objective	Participants / stakeholder groups	Key outcomes	Evidence
24 Sep 2026	Inception meeting	Present technical scope, workstreams and decisions to close	Not recorded	Scope brief prepared; no minutes or attendance record available	Technical scope brief
2 Oct 2026	Written questions to Primero	Obtain API, environment, authentication, test and UAT information	Nelson Adagi to Isaiah Amutavi Osotsi	API reference link; SSO preferred; 60-minute tokens; API credentials set up by Primero; no rate limits; child-journey test data suggested	[TR1] (reply date not supplied)
8 Oct 2026	Session 1 — Business and workflow	Confirm workflow and referral data	BITZ (Nelson, Peter, Marlon); Primero (Isaiah); Ministry IT (Hopolang); Child Helpline social workers, supervisors and a research analyst; CPIMS focal person (Tiisetso Mohlabi); UNICEF absent	Manual trigger; perpetrator details included; ABR optional; prank and follow-up updates pushed to CPIMS+;	Session 1 minutes

Date	Event	Objective	Participants / stakeholder groups	Key outcomes	Evidence
				emergency phone path retained	
9 Oct 2026	Session 2 — Technical architecture	Technical deep dive and CPIMS+ demonstration	BITZ; Primero; Ministry IT; Ministry focal person (Tiisetso Mohlabi)	Service-based routing revision; case plan not shared; status gap identified; audit modelled on CPIMS+ log; field-mapping started; D-001 – D-004	Session 2 minutes
14 Oct 2026	Technical follow-up call	Review outstanding requirements checklist	Planned smaller technical group	No minutes or outcome recorded	Meeting invitation
26 Oct 2026	Session 3 — Closing requirements	Close open business and technical items	UNICEF (chair); Ministry IT and child protection; Child Helpline; Primero; BITZ. Twelve participant entries are recorded in the attendance export [ATT3]; participant roles and frontline helpline representation are to be confirmed	D-005 – D-018; actions A-013 – A-025	Session 3 minutes; action tracker; [ATT3]; [TX3]

Figure 2 — Inception timeline.



3.3 Evolution of positions during inception

Several positions changed. Session 3 governs where it expressly updates an earlier position.

Topic	Earlier position	Current position
Routing	Session 1: helpline refers directly to the district supervisor	D-008: agent selects a CPIMS+ service; the Ministry distributes

Topic	Earlier position	Current position
Feedback	Session 1: one-way follow-up push; return of status not committed	D-007: two-way updates for referred cases
Anonymity	Session 2: anonymous reports cannot be referred	D-010 / D-011: anonymous reporter allowed; child identifying details required
ABR ID	Session 1: optional; checks deferred	D-005 / D-014: optional; the supplied NICR number format will be used for the first checks
Attachments	Session 3 facilitation deck: deferred unless approved	D-012: included in the field set
Narrative	Required; timing not stated	D-013: mandatory at referral, not at first save
UNICEF participation	Sessions 1–2: absent; escalation raised (A-012)	Session 3: UNICEF chaired and attended; approval route still to confirm

3.4 Limitations of the inception evidence

- No minutes exist for the 24 September inception meeting or the 14 October technical call.
- Frontline Child Helpline representation requires confirmation against [ATT3]; the minutes describe a Child Protection Services representative. Attendance names alone do not establish roles.
- The September 2019 SOP has been received [SOP1]; its current applicability is pending. Separate pathways, privacy materials, Berea use cases and the configured service list remain outstanding.
- The unified national referral form, recorded as shared under A-004, has not yet been made available to BITZ.
- The annotated requirements checklist has no recorded author or date; its remarks are treated as positions for confirmation.
- No OpenCHS application-code or live-deployment audit was performed for this report.

3.5 Findings from documents received after Session 3

The September 2019 *HIV-Sensitive Case Management Manual — Standard Operating Procedures for Ministry of Social Development* [SOP1] changes A-014 from awaiting receipt to **Received; current applicability and mapping review pending**. Hopolang / Ministry must confirm that this is the applicable edition and identify subsequent amendments; BITZ must document the resulting workflow and mapping review. The separate national referral pathways (A-013) and privacy materials (A-017) remain outstanding.

Form 4 contains the Child Protection Referral Form and a return receipt. It supplies candidate mapping inputs for child identity, file number, referral date, referring and receiving organisations, service needs, referral reason, accompanying documents, expected results, receiving person and response actions. These must be compared with the agreed D-012 field set and configured CPIMS+ definitions; they do not automatically add fields to the approved scope or establish CPIMS+ service codes. The receipt supports review of human acknowledgement and follow-up under D-016, separately from API delivery acknowledgement. It does not itself define technical acceptance/rejection states. Whether it is the unified national referral form referenced under A-004 remains to be confirmed.

The SOP guiding principles address informed consent/assent and confidentiality; its confidentiality guidance also describes exceptions including emergencies. This provides country procedural evidence for DATA-012 and R-011. Ministry, advised by UNICEF, must confirm the applicable disclosure basis, consent capture, authorised exceptions and return-sharing only necessary information; Primero must confirm configured consent and override controls.

The 2019 manual is treated as procedural guidance rather than a statement of current law, and consent was not agreed as an integration rule during the sessions.

UAT preparation must include a Form 4 mapping review, human receipt and response scenarios, and Ministry-approved consent/exception scenarios. Business response times in the SOP must be reviewed separately from technical retry and delivery targets; no API service-level target is inferred from the manual.

The Session 3 transcript [TX3] corroborates the recorded decisions on manual referral, two-way updates, anonymous reporters, identifiable child details, the optional ABR ID and the requirement for a narrative at referral. It does not alter decisions D-005 to D-018. As an automated transcript, it is read alongside the approved minutes.

The attendance export [ATT3] records 12 participant entries. Attendance records do not confirm job roles or signatory authority. UNICEF is requested to confirm the owners of actions A-017 and A-019 and the contacts named in the minutes (P-03), and the Child Helpline to confirm its frontline representation.

Reference	Source
[SOP1]	September 2019 case-management SOP and tools
[TX3]	Session 3 meeting transcript
[ATT3]	Session 3 attendance export

3.6 Requirements baseline and remaining gaps

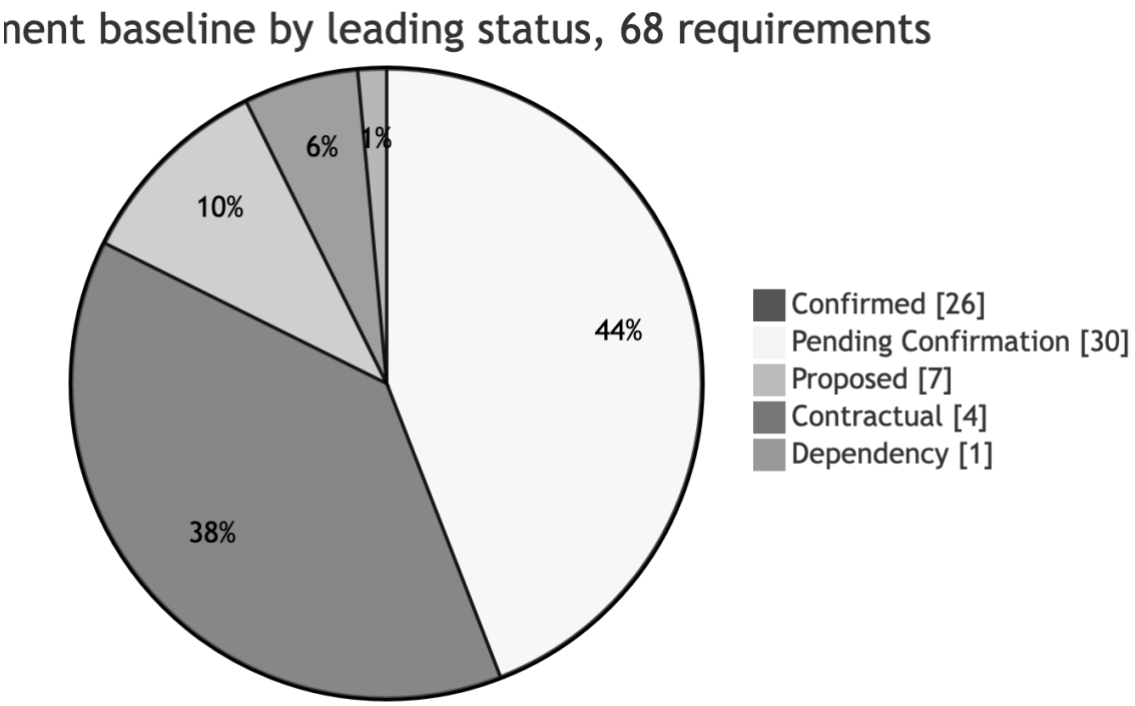
This section summarises the requirement baseline. The full matrix, with sources and acceptance criteria for every requirement, is in **Annex A**. Requirement IDs reuse the completion inventory (GOV, WF, RT, DATA, API, OPS, UAT). New identifiers introduced in this report (FN, SEC, AUD and OPS-008 onward) are **proposed additions** to the inventory.

Priority values are BITZ proposals for governance confirmation. Where a requirement derives from the contract TOR or a Session 3 decision, “Must” reflects that source; it does not imply that priorities have been approved.

Figure 3 — Requirement baseline by status. Requirements are counted by the leading status in Annex J.1–J.8. For example, “Confirmed; mapping Pending Confirmation” is counted as Confirmed. Confirmed means the business decision is recorded in the minutes; its detailed design may still be open.

Section	Confirmed	Contractual	Proposed	Pending Confirmation	Dependency	Total
Annex J.1 Business and workflow	6	0	0	6	0	12
Annex J.2 Functional	10	2	0	0	0	12
Annex J.3 Data and fields	3	0	1	9	0	13
Annex J.4 Integration	2	0	0	6	0	8
Annex J.5 Security and privacy	1	0	4	2	1	8
Annex J.6 Operational and reliability	0	1	1	3	0	5

Section	Confirmed	Contractual	Proposed	Pending Confirmation	Dependency	Total
Annex J.7 Audit and monitoring	2	1	1	0	0	4
Annex J.8 UAT and acceptance	2	0	0	4	0	6
Total	**26**	**4**	**7**	**30**	**1**	**68**



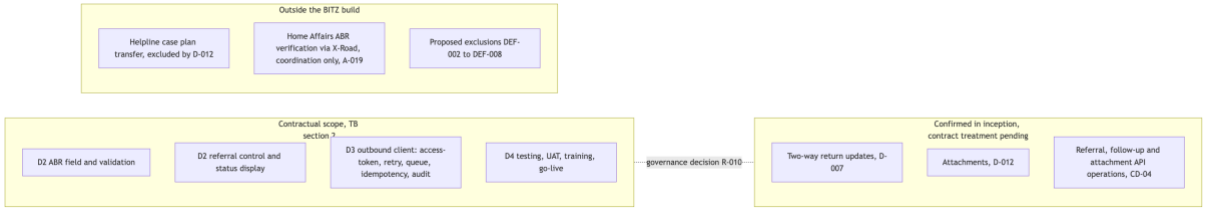
The functional requirements, which drive D2, are almost entirely confirmed. Most open items are in data mapping, integration and security, and they depend on the configured Primero environment and the country documents.

4 Scope and boundaries

4.1 Scope classification

Category	Items	Status
Original contractual scope	Outbound referral client; ABR field and checks; referral control and status display; access tokens, retry, queuing, duplicate prevention, audit; testing, UAT, training, so-live	Contractual
Requirements confirmed during inception	Manual agent trigger (D-006); service selection with Ministry distribution (D-008); anonymous reporter with identified child (D-010, D-011); field set incl. attachments, excl. case plan (D-012); narrative at referral (D-013); NICR ABR format (D-014); two acknowledgements (D-016); API spec basis (D-017); global demonstration system fallback (D-018); Ministry-led UAT (D-009)	Confirmed
Requirements confirmed during inception that extend contract wording	Two-way return updates (D-007); attachments transfer (D-012); referral and follow-up operations beyond case creation	Confirmed requirement; contract treatment Pending Confirmation (section 4.2)
Proposed technical solutions	Integration architecture, duplicate prevention key, retry schedule, return-update options, audit design (sections 6 and 8)	Proposed
Coordination outside BITZ build scope	Participation in Primero/X-Road sessions on Home Affairs ABR verification (A-019, A-020)	Confirmed participation; implementation Proposed exclusion (DEF-009)
Excluded	Helpline case plan transfer (D-012)	Excluded (Confirmed)
Proposed exclusions	Call recordings and large media not expressly agreed (DEF-002); advanced Primero lookup administration (DEF-004); generic integration marketplace (DEF-005); major OpenCHS workflow redesign (DEF-006); non-essential analytics (DEF-007); automated production deployment pipeline (DEF-008)	Proposed exclusion

Figure 4 — Scope boundary. Sources: [TB section 2], [M3], completion inventory section H.



4.2 Two-way integration

Aspect	Position
Original contractual position	Workstream 3 is an *outbound* REST API client [TB sections 2 and 5]. The brief asked inception to confirm whether status information must be written back, and said any bi-directional requirement must be explicitly confirmed [TB section 9].
Updated business requirement	Session 3 D-007: the integration is two-way; updates to a referred case in CPIMS+ flow back to the helpline. Confirmed by the Child Helpline representative, the Ministry and UNICEF.

Aspect	Position
Technical implications	Inbound endpoint or polling component; authentication of inbound calls; agreed status vocabulary; ordering, duplicate and conflict handling; reconciliation because standard Primero webhook delivery is not retried (Annex B.5); OpenCHS display of returned updates.
Security and data-sharing implications	New flow of Ministry case data into OpenCHS; minimisation of returned fields; access control on returned data; retention; lawful basis for the return direction (section 7).
Testing implications	Additional integration, security and UAT scenarios (Annex H RC-05, F-12, S-04).
Schedule and resource implications	Additional design, build and test effort not reflected in the D3 milestone. No estimate has been prepared; BITZ proposes to provide one after the mechanism is agreed (RT-007).
Related extensions	Attachments (D-012) and the referral, update and attachment operations beyond POST /api/v2/cases (CD-04).
Formal decision required	Whether return updates and attachments are (a) absorbed within D3 and D4, (b) handled by contract amendment or variation, or (c) phased; and what acceptance evidence applies.
Responsible governance authority	UNICEF as contracting party, with the Ministry as business owner. Named approver Pending Confirmation (GOV-003).

Agreement on a business requirement in Session 3 is not approval of a contractual variation. Until governance decides, BITZ will design for two-way updates but will report return-update work separately from the contracted outbound D3 scope.

5 Business requirements and workflow

This section sets out the business requirements agreed during inception for referring cases from Child Helpline 116 to the Ministry through Primero CPIMS+. It describes the agreed referral lifecycle, the business rules that apply at each stage and the matters still to be confirmed. Requirement identifiers refer to the detailed matrix in Annex A; decision identifiers refer to the decision register in Annex E.

Table 5.1 — Agreed referral lifecycle

Step	Activity	Reference
1	The agent receives the call, records the contact and saves the helpline case. The narrative need not be complete at first save.	D-013
2	The handling agent decides to refer and selects an approved CPIMS+ service, optionally escalating to a supervisor first.	D-006, D-008; [M1 section 6]
3	OpenCHS validates the narrative and the required child-identifying information, allowing an anonymous reporter and an absent ABR ID.	D-005, D-010, D-011
4	The integration submits the agreed data and records a system acknowledgement, or a failure requiring recovery.	D-016
5	The receiving party accepts or rejects the referral through the CPIMS+ workflow. System receipt does not constitute human acceptance.	D-016
6	Ministry users distribute and assign the case internally; the helpline does not select a caseworker.	D-008
7	OpenCHS sends follow-up information against the linked case; CPIMS+ returns approved status and intervention updates.	D-007, D-012
8	Closure, correction, cancellation and re-referral retain the case linkage and an auditable history; detailed rules are pending.	WF-006

Urgent cases continue to be telephoned to the police or first responders in parallel with the referral (WF-007). The complete lifecycle, with the native CPIMS+ referral mechanism and reliability design, is detailed in Annex D.

5.1 Referral initiation and intake rules

The agreed workflow begins with a helpline agent manually initiating a referral. Saving a case does not send it. The referral must include a narrative and identifying child details, with minimum fields and emergency exceptions still to be confirmed. Reporters may remain anonymous and missing ABR IDs must not block referral. Urgent cases retain the parallel emergency telephone route.

Table 5.2 — Business rules at referral

Rule	Requirement	Status	Reference
Referral initiation	The handling agent initiates the referral manually; saving a case never sends it.	Confirmed	D-006; WF-002
Case narrative	A narrative is required at referral, not at first save.	Confirmed	D-013; FN-004
Child identity	Identifying details of the child are required at referral.	Confirmed; fields Pending	D-011; FN-005
Reporter anonymity	An anonymous reporter is permitted and anonymity is preserved in the data sent.	Confirmed	D-010; WF-009
ABR ID	Optional; a blank value never blocks referral; structural validation applies when entered.	Confirmed	D-005, D-014; FN-001, FN-002

Rule	Requirement	Status	Reference
Emergency referral	A minimum-information referral is permitted for emergencies.	Pending Confirmation	D-003; WF-004
Urgent cases	The parallel telephone route to the police and first responders continues.	Confirmed	WF-007
Supervisor escalation	Optional escalation to a supervisor; any mandatory-escalation rules are to be defined.	Pending Confirmation	WF-003

5.2 Case distribution and status tracking

The agent selects an approved CPIMS+ service. Ministry users distribute the case; the helpline does not choose a caseworker. Technical receipt, human acceptance or rejection, and subsequent case progress must remain distinct. Two-way updates apply to referred cases, subject to the return-data contract and governance treatment of the scope change.

In CPIMS+, the recipient sees an incoming referral as pending and either accepts it, which opens the case file, or rejects it with a reason. Assignment to a caseworker by a district manager is a separate, later step. The helpline requires visibility of acceptance, service completion and closure, but not of internal assignment [M2 section 4].

The helpline-facing status vocabulary is to be derived from the standard Primero referral statuses rather than defined separately. The proposed states are Queued, Received, Accepted, Rejected, In progress, Completed and Withdrawn, with Needs action for validation errors or delivery failures requiring intervention (Annex D.3; RT-006, Pending Confirmation).

5.3 OpenCHS interface and scope limits

OpenCHS must show referral status and messages explaining missing or incorrect information, prevent duplicate submission, and preserve links for follow-up updates. The helpline case plan is excluded. Supervisor escalation, partial referrals, correction, cancellation and re-referral rules remain pending.

Follow-up information, including a later finding that a call was a prank, is sent as an update to the linked CPIMS+ case rather than as a new referral (WF-010; [M1 section 7]).

Table 5.3 — Business requirements pending confirmation

Matter	Requirement	Status	Reference
Supervisor escalation	Rules for optional and any mandatory supervisor escalation	Pending Confirmation	WF-003
Emergency referral	Approved minimum dataset for emergency referrals	Pending Confirmation	WF-004
Partial referral	Rule for referring with partial information and adding information later	Pending Confirmation	WF-005
Correction and cancellation	Correction, cancellation, withdrawal and re-referral, with history	Pending Confirmation	WF-006
Service list	Approved list of CPIMS+ services available for referral	Dependency	RT-001; A-018
Routing configuration	Configured recipients, reasons and assignment permissions	Pending Confirmation	RT-003

Matter	Requirement	Status	Reference
Status vocabulary	Intermediate statuses shown to helpline staff	Pending Confirmation	RT-006
Return updates	Content and mechanism of follow-up and return updates	Pending Confirmation	RT-007
Child identity fields	Minimum child-identification fields and their mapping	Pending Confirmation	DATA-005

5.4 Detailed requirements

Detailed business and functional records are retained in Annex A.1–A.2 and Annex J.1–J.2. Annex D contains the full referral lifecycle, native CPIMS+ workflow, proposed status vocabulary and reliability detail. Section 7 addresses the information being shared and its safeguards.

6 Proposed technical approach

This section sets out the proposed technical approach for exchanging referrals and case updates between OpenCHS and Primero CPIMS+, giving helpline staff visibility of each referral after it is made. The approach will be validated against the Lesotho deployments of both systems before the detailed design is approved. The detailed design, architecture diagrams and security specifications are provided in Annex K.

6.1 Design principles

A helpline agent must choose to send a referral; saving a case must not send it automatically. The system should prevent the same referral from creating duplicate cases, keep unsuccessful referrals for follow-up, and show staff when action is needed. Only agreed information should be shared. A record of each exchange should show what happened without unnecessarily copying sensitive child information.

6.2 Integration architecture overview

OpenCHS will send the referral to Primero CPIMS+. The Ministry will receive and distribute the case through CPIMS+. Updates on the referred case will return to OpenCHS. The helpline will continue to telephone emergency responders when urgent action is needed.

The Ministry and technical teams must choose whether to connect the systems directly or use OpenFN, an open-source integration platform (D-015). X-Road concerns the separate Home Affairs checks; it is not the agreed route for referrals between OpenCHS and CPIMS+.

6.3 Referral transaction flow

1. The agent selects the service needed and checks the referral information.
2. OpenCHS checks that the required child details and case description are present.
3. The referral is sent securely to the Ministry through CPIMS+.
4. OpenCHS records whether the system received it. This does not mean a person has accepted the referral.
5. A Ministry user accepts or rejects it, giving a reason where required, and the Ministry assigns the work internally.
6. Agreed progress and service updates return to the linked helpline case.

The precise steps the software will follow and the information returned still need confirmation (API-003, RT-007).

6.4 Error handling and recovery

Under the proposed approach, referrals that cannot be delivered are queued and retried automatically. Repeated attempts should not create duplicate cases. Staff should be able to see failures, correct information where necessary and obtain support. The teams must agree how often to try again, when to alert staff and how to check that both systems hold consistent information. These safeguards still require testing.

6.5 Integration responsibilities

Part of the work	Responsibility and confirmation needed
Helpline form and referral button	BITZ must check and update OpenCHS without disrupting existing work

Part of the work	Responsibility and confirmation needed
Referral delivery	BITZ must implement delivery, failure handling and duplicate prevention
Receiving and assigning referrals	Primero and the Ministry must confirm services, authorised recipients and permissions
Returning progress updates	Both technical teams must agree how updates return and which information is included
Monitoring and support	Named owners must monitor failures, keep records and respond to incidents

The design will be verified against both deployed systems before approval. Earlier technical notes proposed automatic referral on saving a case; this conflicts with decision D-006 and is excluded from the design.

6.6 Development and test environments

Setting	Purpose and current evidence
BITZ development system	Used for component development and unit testing; end-to-end behaviour is verified in the test environments
OpenCHS test system	Being prepared in the recorded baseline; access evidence remains pending
Lesotho CPIMS+ test system	Preferred for testing country-specific forms and procedures; readiness checklist and access remain outstanding
Primero global demonstration system	Agreed fallback if the Lesotho test system is unavailable; country-specific tests must later be repeated
Production systems	Use begins only after testing, training and approval; access, support and recovery arrangements still need confirmation

The status shown reflects the position recorded at 28 October 2026.

6.7 Reliability and audit logging

Each exchange should have a traceable reference, result and history of delivery attempts. Staff need clear messages when a referral is waiting, needs correction or has failed. Detailed connection requirements remain in Annex B and Annex J.4, J.6 and J.7. Annex K preserves the technical options, which remain proposals until confirmed and tested.

7 Data and safeguarding requirements

7.1 Data requirements and mapping approach

The agreed referral information set includes child details, case category and narrative, reporter contact where available, perpetrator details, attachments and follow-up updates. The ABR ID remains optional and the helpline case plan is excluded. Configured identifiers, minimum fields, code lists, attachment controls and the representation of consent remain pending. Annex C retains the mapping baseline and ABR rules; Annex J.3 retains detailed data requirements.

The September 2019 SOP provides Form 4 and confidentiality guidance, which will inform the mapping and safeguarding design. The Ministry is requested to confirm its current applicability and the approved disclosure and exception rules. Section 3.5 records the evidence and its implications.

7.2 Legal and policy position

The sources differ on the applicable law. The supplied contract requires compliance and confirmation of data-sharing agreements but does not resolve which legal instrument is in force. In Session 3, UNICEF noted that such legislation is under development; the Session 3 action register refers to the Data Protection Act 2011; the annotated checklist refers to the Lesotho Data Protection Act 2012. **Determining the applicable legislation and its status is outside the scope of this report.** UNICEF will supply earlier Ministry data-protection work, the data management policy and legal material (A-017); BITZ will complete desk research (A-022). The responsible Ministry and UNICEF advisers must confirm the instrument, year and commencement status.

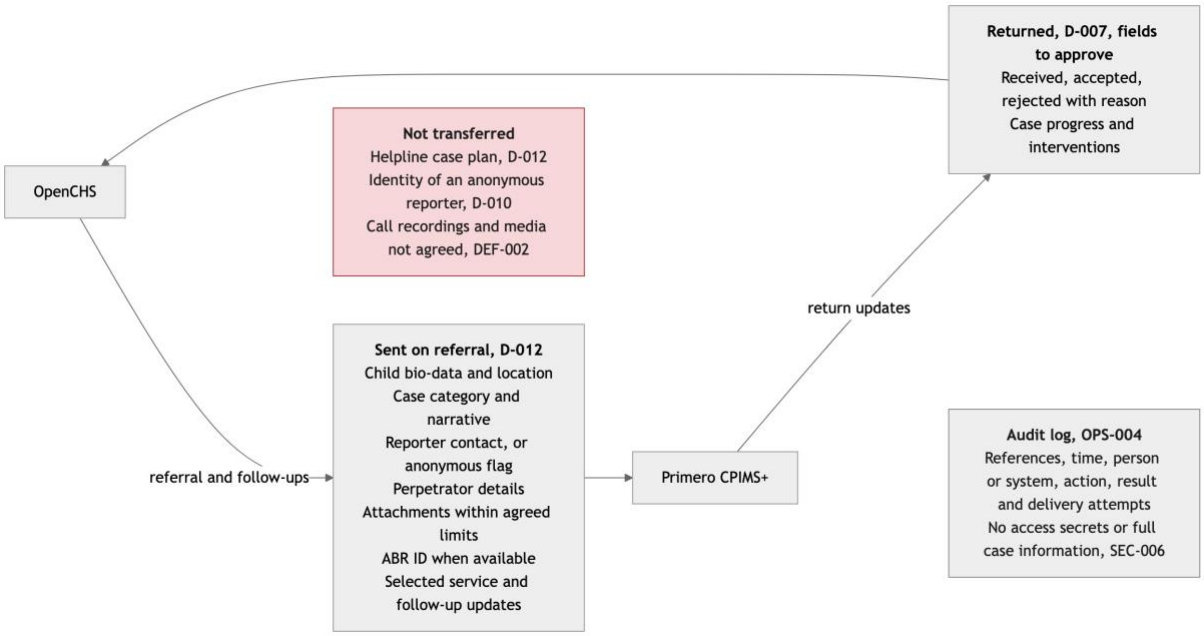
7.3 Information security controls

Control area	Requirement	Status	Owner
Lawful basis	Authorised purposes for sharing child-protection data between the helpline and CPIMS+	Dependency (A-017, A-022)	Ministry / UNICEF
Consent and disclosure	Standard Primero checks consent for disclosure and services at referral unless overridden; no consent capture was discussed in the sessions. Decide how consent or another basis is captured and who may override it	Pending Confirmation (R-011, DATA-012)	Ministry; Primero to confirm configuration
Anonymous reporters	Reporter recorded as anonymous in CPIMS+; anonymity preserved in information sent, logs and return updates	Confirmed (D-010); mapping Pending	BITZ, Primero
Child-sensitive data	Only agreed fields sent; no fictitious names, dates or identifiers created to satisfy checks	Confirmed field set (D-012); rule Proposed	BITZ
Access control	Integration account permitted to set only approved fields; OpenCHS role controls for referral and returned data	Proposed (SEC-003)	Primero; BITZ
Authentication	The systems must prove they are authorised to connect. The method for automatic exchanges still needs agreement; technical options are retained in Annex K	Pending Confirmation (API-002)	Primero; BITZ
Protecting information in transit	Encrypt information as it moves between systems and check the receiving system’s identity. Minimum technical safeguards remain in Annex K	Proposed; controls Pending Confirmation	Primero; infrastructure

Control area	Requirement	Status	Owner
Managing access details	Keep passwords and other access details protected, separate testing from everyday use, and agree who holds and changes them	Pending Confirmation (API-008)	BITZ; Primero
Records of system activity	Keep a record of exchanges without copying access secrets or complete case information; approve who can read these records and how long they are kept	Pending Confirmation (OPS-004, OPS-005)	Security owner
Sharing only necessary information	Case plan excluded; return updates limited to referred cases and approved fields	Confirmed (D-012); bound Proposed (SEC-008)	Ministry; BITZ
Retention	Retention of queued information sent, logs and attachments	Pending Confirmation	Data protection owner
Incident handling	Contacts, escalation and notification duties for data incidents and service interruptions	Pending Confirmation (OPS-007)	Ministry; UNICEF
Secure test data	Approved fictional child-case examples; no real child records copied into test environments	Proposed (SEC-007); approval Pending	UAT lead; privacy owner
Attachments	File types, size limits, malware scanning, field permissions, retention; no call recordings unless approved	Pending Confirmation (DATA-010)	Ministry; Primero; BITZ
Return updates	Authenticated channel; minimised fields; webhook role limits fields where used	Pending Confirmation (RT-007)	Primero; BITZ

7.4 Data flows and classification

Figure 5 — Data flow and classification. Sources: D-007, D-010, D-012, OPS-004. Which fields are returned still needs approval (RT-007).



Data	Direction	Sensitivity (proposed)	Control
Child identity, location, contact	OpenCHS → CPIMS+	High (child-identifying)	Agreed fields only; encrypted connection; account limited to necessary permissions

Data	Direction	Sensitivity (proposed)	Control
Narrative, case category, perpetrator details	OpenCHS → CPIMS+	High (child-protection detail)	Not written to logs; role-based access
Reporter contact	OpenCHS → CPIMS+ when given	High	Anonymous flag respected (D-010)
Attachments	OpenCHS → CPIMS+	High	Type, size and scanning rules (DATA-010)
ABR ID	OpenCHS → CPIMS+ when available	High (national identifier)	Structural checks; stored as text
Referral status and interventions	CPIMS+ → OpenCHS	High	Approved fields only; authenticated channel
Records of exchanges	Both systems	Medium	Retention and access approval (OPS-004, OPS-005)
Helpline case plan	Not transferred	—	Excluded from information sent (D-012)

7.5 Matters requiring UNICEF or Ministry confirmation

The supplied contract explicitly adds confirmation of data-sharing agreements among MSD, MoH, MoE and UNICEF [CT1 section 5]. The status of these agreements, and the authority for sharing real case data, must be confirmed before deployment; an agreement still in preparation is not sufficient.

1. Applicable data protection instrument and its status.
2. Lawful basis and authorised purposes for each direction of exchange.
3. Consent capture and override authority.
4. Retention periods for integration data, logs and attachments.
5. Incident notification obligations and contacts.
6. Approval of the fictional records for testing and prohibition of real records in test.
7. Named data protection/security approver.

7.6 Safeguarding requirements

Reporter anonymity must be preserved in both directions. Integration permissions, transport security, secret management, log redaction, synthetic test data and restrictions on returned fields require agreed controls. Annex J.5 and Annex A.5 retain the detailed security requirements and their confirmation status.

8 Implementation plan

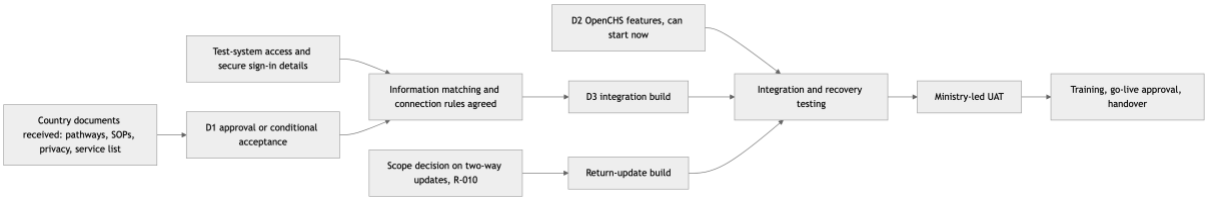
8.1 Work plan and delivery phases

The work plan is organised into twelve phases, each mapped to the contractual deliverables. Outputs not named in the contract are marked as proposed supporting outputs.

8.1 Work plan and delivery phases								
#	Phase	Objective	Principal activities	Expected outputs	Responsible	Dependencies	Completion / acceptance criteria	Deliverable
1	Requirements confirmation	Close the inception baseline	Confirm received SOP applicability; receive pathways and privacy material; resolve minimum information, consent, service interruption rules; record approvals	Approved D1 report; updated inventory and registers	BITZ (R); Ministry, UNICEF (approval; named approvers Pending)	A-013, A-014, A-017, A-021; GOV-001	Written D1 approval or conditional acceptance with outstanding items with an agreed owner, deadline and effect on delivery	D1
2	Detailed solution design	Agree the technical interface	Sequence design; duplicate prevention; method for returning progress updates; agreed status vocabulary; error handling	Integration design specification and interface agreement (proposed supporting output)	BITZ, Primero	Configured environment; GOV-004; API-002	Design reviewed by named technical authorities	D1 / D3
3	Environment preparation	Usable test environments on both sides	OpenCHS test/UAT; Primero test system or global demonstration system; credentials package; secret storage	Access package; environment matrix	BITZ (OpenCHS); Primero (CPIMS+); Ministry/UNICEF (checklist)	A-015, A-016	Successful authorised exchange of test information recorded	D1
4	System setup and development (OpenCHS)	Build D2 features	ABR field and checks; referral control; narrative and identity rules; status display	Demonstrable features; unit tests	BITZ	Minimum information (WF-004); service list (A-018)	Demonstration and unit-test evidence against D2 criteria	D2
5	API integration	Build the integration client and return path	Authentication; offline queue; retries; duplicate prevention; referrals, updates and attachments; records of exchanges	Integration service; contract tests	BITZ	Phases 2–3; R-010 decision for return path	Contract tests pass; live calls against test environment evidenced	D3
6	Security and functional testing	Verify controls	Check access permissions, protection of information in transit, activity records and limits on information shared	Test report (proposed supporting output)	BITZ; security owner (Pending)	SEC requirements approved	No open Critical/High security faults	D3 / D4
7	Integration testing	Verify end-to-end behaviour	End-to-end referrals; failures; service interruptions; duplicates; out-of-order updates; regression testing	Integration test evidence	BITZ, Primero	Configured environment	Annex H integration scenarios pass	D4
8	UAT	Ministry-led acceptance	UAT plan; fictional test records; execution; review and prioritisation of faults; retest	Signed UAT report	Ministry (lead, D-009); BITZ (support, test data per checklist, Pending Confirmation); Child Helpline; Primero	UAT-002, UAT-003, UAT-007	Section 10 proposed conditions for completion met and signed	D4
9	Training and knowledge transfer	Operational and technical capability	Operator/social-worker and IT training	Materials; attendance records	BITZ; Ministry (participants)	Stable UAT build	Training evidence submitted	D4
10	Deployment preparation	Production readiness	Production endpoints, API credentials, rollback, monitoring, support rota	Readiness checklist; rollback plan	BITZ; infrastructure owners; Primero	API-008; OPS-006	Go/no-go decision recorded	D4
11	So-live support	Stabilise in production	Hypercare (post-launch support), monitoring, incident handling	Hypercare log (proposed supporting output)	BITZ; Ministry support owners	So-live approval	Agreed hypercare exit conditions met	D4
12	Handover	Transfer to operations	Technical handover report, user manual, administrator guide	Handover pack	BITZ	Phases 9–11	Handover accepted by named owner	D4

8.2 Implementation and dependency flow

Figure 6 — *Proposed sequencing – not an approved schedule*



8.3 Original contractual timetable

Figure 7 — Contractual timetable. The supplied contract covers 1 July 2026 to 28 February 2027 [CT1 sections 3–4]. These are the original contractual targets. The milestone months group several tasks; an October software task can form part of the November acceptance milestone.

Period	Required work	Evidence or approval
July–August 2026	Inception, design, information mapping, system preparation, repository access and ABR specification	D1 report, design, mapping and setup evidence; August Item 10 milestone
September 2026	Optional ABR field and required number checks	Demonstration and test evidence; part of Item 20
October 2026	Referral checkbox and initial connection to Primero	Demonstration and software tests; October Item 20 milestone; initial connection contributes to Item 30
November 2026	Complete connection, activity records, linked Primero case reference and operator alerts	Item 30 evidence and review; usable Primero access needed for subsequent live tests
December 2026	Test individual parts and use a simulated receiving system	Test and coverage reports; simulated tests supplement but do not replace live integration testing
January 2027	Test both systems together, check transferred information, conduct user testing and resolve faults to UNICEF satisfaction	Integration results and UNICEF-executed user-testing sign-off; Ministry leads user testing under D-009
February 2027	Deploy, check the released system, train staff in Maseru and complete handover	UNICEF launch confirmation, attendance, agreed English/Sesotho materials, manuals, technical report, administrator guide and Service Provider assessment; Item 40
Ninety days after launch	Free maintenance support	Named support owners and recorded support period; end date follows actual launch and requires confirmation



Bars show the contractual month windows, not separately agreed day-level deadlines. Allow UNICEF five working days to review each submission, plus time for corrections and any further review [CT1 section 6]. Final submission cannot be left to 28 February if acceptance is required by that date. Exact submission and review dates require agreement using the applicable working calendar.

8.4 Current status against the baseline

Deliverable	Target	Status as at 27 Oct 2026	Evidence
D1	Aug 2026	Target month passed. Requirements substantially defined; report in draft; mapping, environment evidence and approval outstanding	This report; inventory
D2	Oct 2026	Design inputs largely available (ABR format, manual trigger, narrative rule). No build or test evidence submitted to date	—
D3	Nov 2026	Blocked on environment access, authentication, configured mapping and the scope decision on return updates	A-016, A-023, R-010

Deliverable	Target	Status as at 27 Oct 2026	Evidence
D4	Feb 2027	Not started; Ministry-led testing agreed (D-009); UNICEF contractual sign-off role supplied [CT1]; no test results yet	—

Whether the D1 milestone has been invoiced, accepted or formally extended is to be confirmed through project governance (R-009).

8.5 Outstanding work and critical dependencies

[CT1 section 5] requires UNICEF to confirm data-sharing agreements involving MSD, MoH, MoE and UNICEF. Real case data will not be exchanged until the necessary agreements and authorisations are in place. Home Affairs must also supply the ABR check-digit specification and at least 10 valid examples. Although the field is optional and must not block a referral when left blank, its validation rules must still be agreed and tested.

Outstanding work	Critical dependency	Owner of dependency
Field-mapping baseline	Configured field definitions and lookups after test system set-up	Isaiah / Primero
Service routing design	CPIMS+ service list (A-018) and service-to-recipient configuration	Isaiah; Ministry
Referral rules and exceptions	SOPs (A-014), pathways (A-013), unified referral form	Hopolang; Rantsala
Authentication and test calls	Environment access (A-016)	Isaiah
Privacy and consent design	Privacy materials (A-017); desk research (A-022)	UNICEF; Nelson
Return-update build	Scope decision (R-010); mechanism agreement (RT-007)	UNICEF / governance; Primero
Service interruption handling	A-021 working session	Nelson, Peter, Isaiah
D1 approval	Named approvers (GOV-001, GOV-003)	UNICEF; Ministry

8.6 Work that can proceed while test system access is pending

Track	Work	Constraint
D2 build	ABR field and checks on the number format; referral control; narrative and identity checks at referral; status display framework	Minimum identity fields remain configurable until WF-004 is decided
D3 foundation	Build the offline queue, retries, duplicate prevention, activity records and authentication, using a simulation of the documented connection	Simulated results supplement, but do not replace, live integration testing
Test design	Annex H converted to test cases; fictional child-case examples drafted	Dataset needs approval (UAT-003)
Governance	Scope, schedule and approver decisions	Governance meeting needed
Training design	Outline operator and IT training	Final content follows UAT build

8.7 Proposed recovery timetable

The following recovery timetable is proposed for review and approval. It retains the original contractual dates and sets out the work required to protect January testing and the February handover. BITZ will confirm the workload and the two-person staffing arrangement before committing to these dates.

Proposed period	Recovery work	Owner	Condition and evidence
Late October–early November 2026	Submit revised D1; confirm governing contract, referral behaviour, ABR rules and two-way scope; review unfinished D2 work	BITZ; UNICEF; Ministry	Submission record; UNICEF review within five working days; written decisions and revised task dates
November 2026	Close access and data-mapping gaps; complete outstanding D2 work and D3 integration	BITZ; Primero; UNICEF facilitating inputs	Usable access and agreed fields; build/test evidence. Any missed October target remains a delay requiring approval
By end November 2026	Verify that the Lesotho test system can support the January test plan	Primero; UNICEF; BITZ	Successful authorised test exchange, required forms and accounts. This reflects the contract's later access assumption and is not an approved extension
December 2026	Complete component tests and simulated exchanges; fix faults; prepare realistic fictional cases and training materials	BITZ; Ministry; Primero	Test coverage report, fault register and approved user-test plan
January 2027	Test the real connection, correct faults, conduct Ministry-led user testing and obtain UNICEF sign-off	BITZ; Primero; Ministry; UNICEF	Actual test results, corrections and signed approval
February 2027	Deploy following approval; deliver in-country training and final handover; obtain final acceptance by 28 February	BITZ; UNICEF; Ministry	Release checks, training attendance, agreed language materials and complete handover pack; allow review and correction time
Actual launch date plus 90 days	Provide free maintenance	BITZ; UNICEF confirms arrangements	Agreed start/end dates and named support contacts; clarify administration beyond February

If access, agreement on extra scope or completion of development cannot support this sequence, BITZ should submit the effect on dates and workload to UNICEF for a decision. BITZ will not reschedule testing or assume a contract extension without UNICEF's written agreement.

Approval checkpoints: G1 — D1 accepted or written conditional treatment agreed; G2 — usable access and approved data mapping; G3 — development complete; G4 — integration tests passed; G5 — user testing signed by UNICEF. The Ministry leads user testing; named approval representatives remain to be confirmed.

8.8 Decisions required to approve the revised schedule

1. Confirm TICAD-2026-001 as the governing contract and explain the earlier contract reference; obtain execution evidence and general terms (CD-02).
2. Record actual D1/D2 progress and approve treatment of missed targets; agree submission dates allowing five working days for UNICEF review and time for corrections.
3. Agree the effort and contract treatment of two-way updates and attachments, and confirm the selected-checkbox/save behaviour (CD-01, CD-03, CD-04).
4. Reconcile August/Week 4 access requirements with the end-November assumption; confirm the week-number anchor and obsolete Week 14 reference (CD-07, CD-08).
5. Confirm access, required fields, official ABR rules and examples, and evidence of data-sharing agreements before real-case deployment.
6. Name the UNICEF user-testing signatory and agree how Ministry endorsement is recorded (CD-09).

7. Confirm February travel, training languages, submission formats and the 90-day maintenance arrangements (CD-10, CD-11).

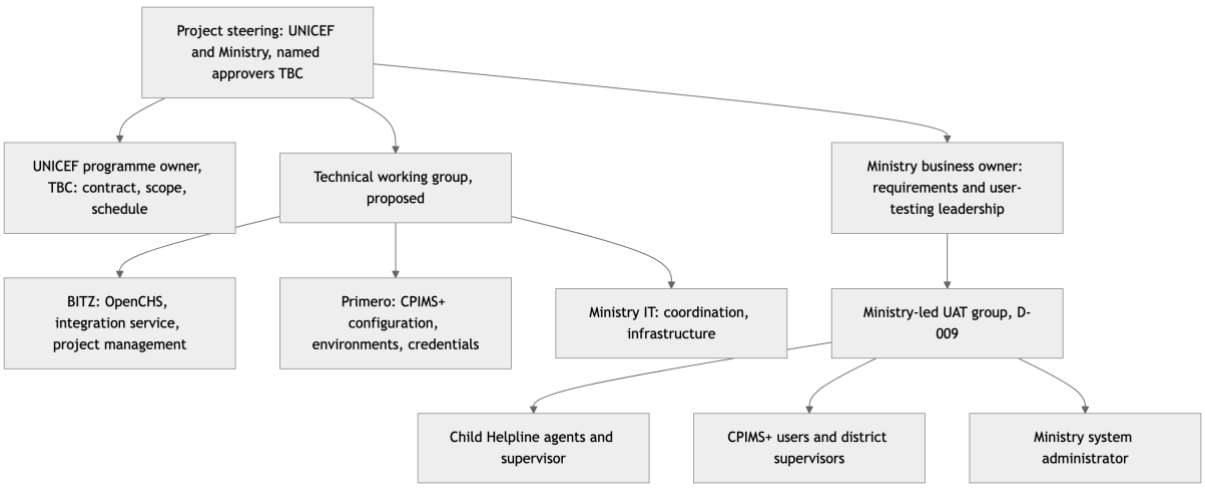
9 Governance and coordination

9.1 Roles

[CT1 section 6] specifies a two-person core team with shared quality assurance. BITZ will confirm named roles, time allocation and replacement arrangements before committing to the recovery plan. The table below lists all parties consulted; inclusion does not imply a full-time delivery role.

Party	Recorded responsibilities	Still to be confirmed
UNICEF Lesotho (Rantsala Sanaha; Ramoletane Lekhanya)	Contracting party; chaired Session 3; pathways (A-013); test system checklist with Primero (A-015); privacy materials (A-017); X-Road invitation (A-019)	Named approver and escalation route (GOV-003); confirmation of action owners (P-03)
Ministry (MGYSD / Ministry of Social Development) — IT (Hopolang Mosena; Melusi, as named in the minutes)	Host and local coordination; SOPs (A-014); Berea use cases (A-020); helpline cover (A-024); permanent focal person (A-002); anonymity rule accepted in Session 3	Business owner and signatory
Ministry — child protection (Tiisetso Mohlabi, CPIMS focal person)	Ministry-level coordination with Primero (Sessions 1–2)	Role in approval
Child Helpline 116	Workflow users; UAT participants (D-009); testing contact (Mary Damposi discussed, not confirmed)	Focal person and leave cover; named testers
Primero (Isaiah Amutavi Osotsi)	Technical coordination; test system/global demonstration system (A-016); service list (A-018); API credential set-up; field definitions after setup	Deployed release; authentication flow; method for returning progress updates
BITZ IT Consulting (Nelson Adagi, Peter Rogendo, Marlon Amunga; James, Executive Director)	Project management and report (A-025); desk research (A-022); service interruption handling (A-021); helpline-side development and connection gateway; authentication and test calls (A-023)	Technical design approval route

Figure 8 — Proposed governance structure. Approver roles are TBC (GOV-001, GOV-003). Only the Ministry-led UAT group rests on a recorded decision (D-009).



9.2 Responsibility assignment (RACI) matrix

Task	UNICEF	Ministry	Child Helpline	Primero	BITZ	Evidence
Inception report preparation	Consulted	Consulted	Consulted	Consulted	Responsible/Accountable	A-025
Requirements approval	TBC	Accountable†	Consulted	Consulted	Responsible	Approver unnamed (GOV-001)
Country documents (pathways, SOPs)	Responsible (A-013)	Responsible (A-014)	Consulted	Informed	Informed	Session 3 actions
Privacy materials and legal basis	Responsible (A-017)	Accountable†	Informed	Consulted	Responsible (A-022)	Session 3 actions
Data mapping	Informed	Accountable† (business)	Consulted	Responsible (definitions)	Responsible (OpenCHS side)	[TR1]; [M2 section 11]
Technical design approval	Informed	TBC	Informed	Consulted/Responsible†	Responsible	No approver named
Primero environment and access	Responsible (checklist, A-015)	Consulted	Informed	Responsible/Accountable (A-016)	Consulted	Session 3 actions
OpenCHS environments	Informed	Informed	Informed	Informed	Responsible/Accountable	[M1 section 9]
Service list and routing configuration	Informed	Accountable†	Consulted	Responsible (A-018)	Consulted	Session 3 actions
Integration build (D2, D3)	Informed	Informed	Consulted	Consulted	Responsible/Accountable	Contract
UAT leadership	Consulted	Responsible/Accountable (D-009)	Responsible	Responsible	Responsible (support)	D-009
UAT test data and scenarios	Informed	Consulted	Consulted	Consulted	Responsible (checklist; Pending Confirmation)	Annotated checklist
UAT sign-off	Accountable; named signatory pending	Leads testing; endorsement to confirm	Consulted	Consulted	Supports and submits evidence	[CT1 section 3.3]; D-009
Issue escalation	Accountable†	Responsible	Responsible	Responsible	Responsible	Proposed
Scope and contract change	Accountable (contracting party; approver TBC)	Consulted	Informed	Consulted	Responsible (raise request)	Contract relationship
Deliverable review and sign-off	TBC	TBC	Informed	Informed	Responsible (submit)	Delivery plan section 4

9.3 Approval mechanisms

Approval	Proposed approver	Evidence required	Status
Requirements baseline (this report)	Ministry business owner and UNICEF programme owner	Written approval or comments resolved	Pending Confirmation
Technical design and interface agreement	OpenCHS and Primero technical authorities	Signed-off design specification	Pending Confirmation

Approval	Proposed approver	Evidence required	Status
Data protection and security controls	Data protection/security authority	Written confirmation of lawful basis and controls	Pending Confirmation
User-testing result	UNICEF executes sign-off; Ministry leads testing	Signed user-testing report	Contract role supplied [CT1 section 3.3]; named signatory pending
Contract variation or schedule rebaseline	UNICEF	Written amendment or approval	Pending Confirmation
So-live	Project steering authority	Go/no-go record	Pending Confirmation

9.4 Meetings, submissions and progress reporting

The following obligations come from the supplied contract [CT1 sections 6–7,10]. Additional working meetings can be agreed as needed.

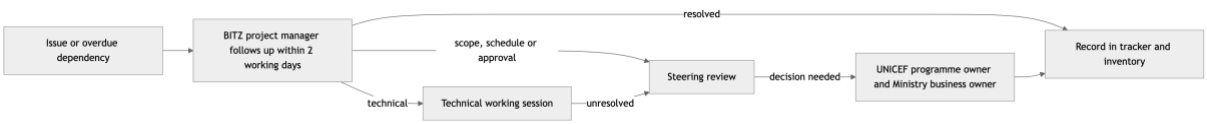
Activity	Timing	Responsibility and evidence
Progress meeting	Weekly; virtual unless otherwise agreed	BITZ and UNICEF/partners review progress, problems and next work
Meeting minutes	Within 24 hours of each weekly meeting	BITZ circulates minutes and actions
Monthly report	Last working day of each month	BITZ reports planned versus actual work, achievement evidence, challenges, lessons and mitigation
Deliverable submission	At each milestone	BITZ uses UNICEF templates, with supporting evidence; plan electronic and hard copy pending CD-11 clarification
Review and approval	Within five working days of submission	UNICEF records comments or approval; schedule corrections and further review where needed
Final technical handover	February 2027	BITZ submits the complete documentation and assessment pack

Weekly progress meetings are separate from any additional technical working sessions. UNICEF's request that a Ministry coordination representative attend every meeting remains applicable [M3 section 1]. The contract specifies two Maseru trips: July/August inception and February training/handover. Arrangements for both trips, including the status of the inception trip, are to be confirmed.

9.5 Escalation path

Proposed. Issue owner → BITZ project manager (two working days) → technical working session or steering review (next session) → UNICEF programme owner and Ministry business owner (for scope, schedule or approval blockers). Overdue dependency actions are escalated at the next steering review.

Figure 9 — Proposed escalation path.

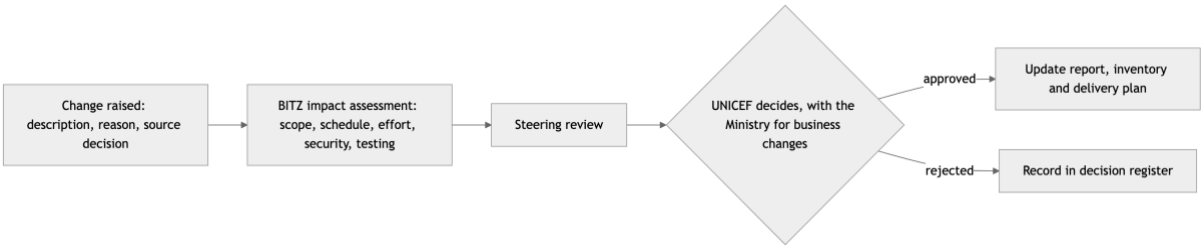


9.6 Change control procedure (Proposed)

1. Any party may raise a change: description, reason, source decision.

2. BITZ assesses impact on scope, schedule, effort, security and testing within an agreed period.
3. The change is reviewed at the steering review.
4. UNICEF (and the Ministry for business changes) decides in writing.
5. Approved changes update this report, the inventory and the delivery plan; rejected changes are recorded in the decision register.

Figure 10 — Proposed change control flow.



9.7 Change log

Change ID	Change	Source	Status
CR-01	Two-way return updates for referred cases	D-007	Requirement Confirmed; contract treatment Pending Confirmation
CR-02	Attachments in the referral information set	D-012	Requirement Confirmed; contract treatment Pending Confirmation
CR-03	Schedule rebaseline following the missed D1 target	R-009	Pending Confirmation
CR-04	Clarify D2/D4 “save-event” wording to manual referral control	CD-03	Proposed

10 Testing and acceptance

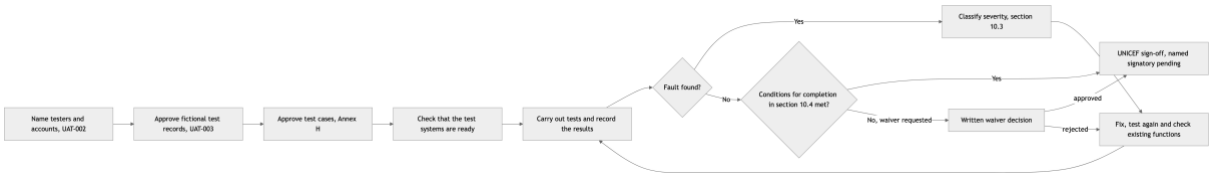
10.1 Test levels and scope

Level	Purpose	Environment	Evidence
Individual features	Check that each new feature works correctly	Development system	Recorded test results
Checks using a simulation	Check the proposed connection against the written specification	Development system	Results from the simulation; end-to-end behaviour is confirmed in integration testing
Both systems working together	Send a referral and follow its progress through both systems	Lesotho test system or approved fallback	Recorded results, screenshots and records of exchanges
Protection of information	Check access, permissions, activity records and secure transfer	Test systems	Security test record
Existing services	Confirm the changes do not disrupt existing OpenCHS functions	Test systems	Recorded regression test results (OPS-008)
Testing by users	Staff confirm that the systems meet agreed working needs	Lesotho test system with country settings	Signed report of staff testing

10.2 User acceptance testing approach (Proposed)

Element	Approach
Leadership	Ministry leads (D-009)
Participants	OpenCHS development team; Child Helpline agents; Ministry system administrator; CPIMS+ users; district supervisors; other Ministry users who receive, distribute, investigate or close referrals (D-009). Child Helpline supervisor per annotated checklist (Pending Confirmation)
Preparation	Named testers and accounts (UAT-002); test cases from Annex H; environment readiness check; UAT briefing
Test data	Approved fictional child-case examples (SEC-007); BITZ prepares per checklist annotation (Pending Confirmation)
Functional scenarios	Annex H, F- and B-series
Security and privacy scenarios	Annex H, S-series
Failure and recovery	Annex H, RC-series
Defect classification	Section 10.3
Retesting	Every fixed Critical/High fault retested by the reporting tester; affected regression tests rerun
Evidence capture	Per test: tester, date, environment, steps, expected vs actual, screenshots or log references, fault ID
Sign-off	UNICEF executes contractual sign-off; Ministry leads testing. Faults must be resolved to UNICEF satisfaction. Any proposed exception needs explicit UNICEF agreement [CT1 section 3.3]

Figure 11 — Proposed UAT process.



10.3 Defect classification (Proposed)

Severity	Definition	So-live rule (Proposed)
Critical	Data loss, duplicate case creation, disclosure of protected data, or referral cannot be sent	None open
High	Required function fails with no acceptable workaround (for example status not returned)	None open unless waived in writing
Medium	Function impaired with a workaround	Documented with fix plan
Low	Cosmetic or minor	Tracked

10.4 Proposed acceptance criteria

The thresholds below are proposed for governance approval. The contract requires all faults to be resolved to UNICEF's satisfaction [CT1 section 3.3]; any waiver proposed below is subject to UNICEF's written agreement.

Criterion	Proposed threshold
Scenario coverage	100% of Annex H scenarios marked Must executed with evidence
Faults	0 open Critical; 0 open High without written waiver
Duplicate prevention	0 duplicate CPIMS+ cases across all retry, resubmission and lost-response tests
Service interruption recovery	100% of referrals queued during a simulated service interruption are delivered or flagged for manual action after restoration; none lost
Audit completeness	100% of transmissions and return updates in the test set have an audit record with all approved fields
Log redaction	No access secrets or complete case descriptions found in the activity records checked
Validation rules	100% of D-005, D-010, D-011, D-013, D-014 test cases behave as specified
Return-update timeliness	Threshold Pending Confirmation; depends on mechanism (RT-007)
Regression testing	No open Critical or High regression defect in existing OpenCHS functions

10.5 Training and handover (D4)

The supplied contract schedules in-country training for February 2027, with materials in English and Sesotho as agreed, and attendance records. Final handover includes the technical report, system architecture and configuration guidance, information-mapping reference, activity-record access and maintenance guidance, user manuals, administrator guide and completed Service Provider assessment form [CT1 sections 3–4].

Operators and social workers: ABR field, referral workflow, messages explaining missing or incorrect information, referral status, returned updates, failure handling and escalation. IT staff: architecture, API configuration, authentication, monitoring, logging, queue management, retries, troubleshooting, environment configuration and support escalation [TB section 6].

10.6 Acceptance requirements and evidence

Ministry-led UAT is agreed, but named testers, safe data, signatories, fault rules and checklist positions still require confirmation. Annex H contains the scenario matrix; Annex J.8 preserves the requirement-level status. Test results, signed acceptance and training evidence will be produced during delivery.

10.7 D1 deliverable completion and acceptance readiness

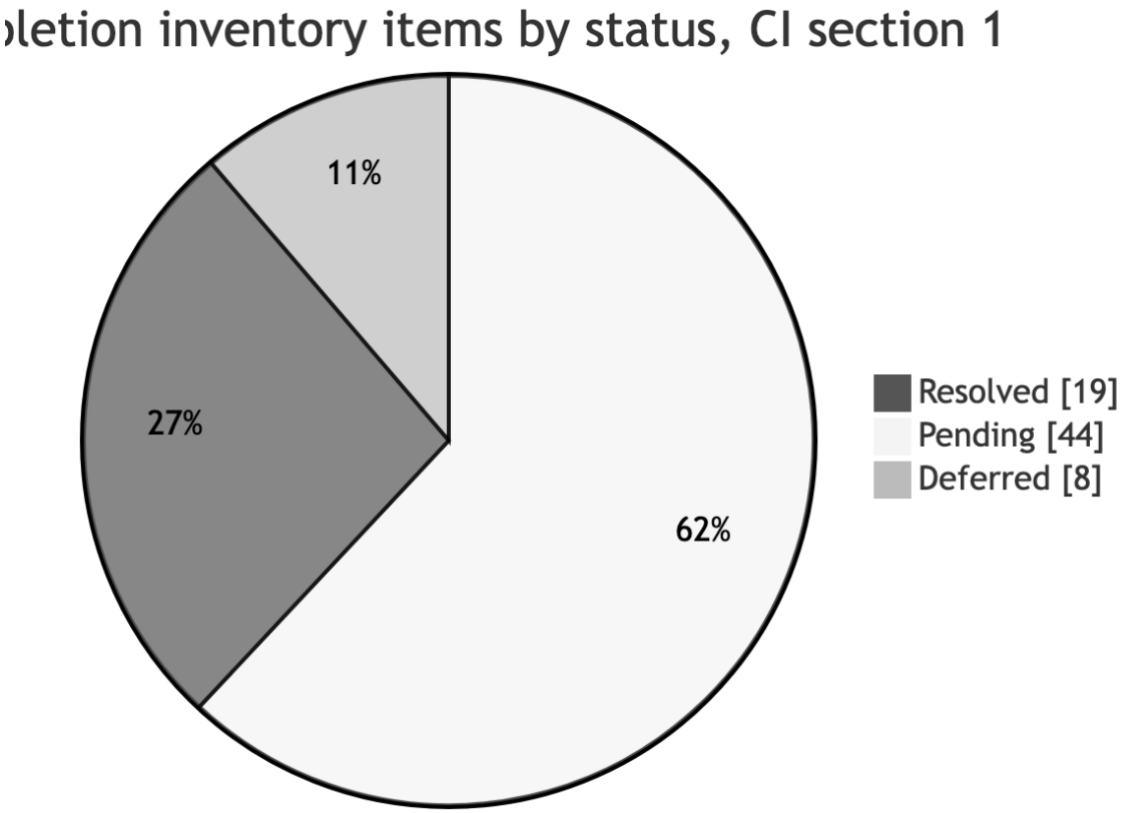
The three columns distinguish **document preparation** (written in a report), **implementation** (built, set up or evidenced) and **formal acceptance** (approved by an authorised person).

Area / Section	Required evidence	Document prepared / Implementation	Formal acceptance	Outstanding work	Owner
Inception consultations	Minutes of sessions with key stakeholders	Completed (Sessions 1–3 minutes)	Not approved	No minutes available for 24 Sep and 14 Oct; Session 3 minutes circulation for correction (S3-N01)	BITZ
Business requirements	Confirmed workflow, rules, field set	Completed (Annex J, Annex A)	Not approved	Minimum information, emergency exceptions, correction/cancellation rules; SOP review	Ministry; BITZ
Technical requirements	System connection, authorised access, protection of information and reliable delivery	Prepared, partly Pending	Not approved	Authentication flow, sequence design, return mechanism, service interruption rules	Primero; BITZ
Integration architecture	Technical design	Proposed (section 6) / Not verified against environments	Not approved	Technical review and testing and sign-off	BITZ; Primero
Data mapping	Approved mapping baseline	Business field set Confirmed; mapping Pending / Configured definitions not received	Not approved	Service list, configured field IDs, code lists	Primero; Ministry
Environment setup	Access and connectivity evidence	Testing-settings table prepared / No access evidence; OpenCHS test/UAT being stood up	Not approved	A-015, A-016, API-005; successful authorised exchange of test information	Primero; BITZ
Work plan	Schedule aligned to contract	Baseline and recovery sequencing prepared	Not approved	Rebaseline decision	Governance
Risk and issue register	Maintained register	Completed (Annex G)	Not approved	Proposed P/I to be validated	BITZ
Decision and action registers	Maintained registers	Completed (Annexes E, F; tracker)	-	Record receipts as they arrive	BITZ
Stakeholder approval	Signed approval or conditional acceptance	-	Not started; approvers unnamed	Name approvers; circulate; resolve comments	UNICEF; Ministry

Figure 12 — Position against delivery gates (27 October 2026). Gates as defined in section 8.7.



Completion inventory status [CI section 1]: 19 Resolved, 44 Pending, 8 Deferred. The inventory includes later implementation, UAT and handover items, so these counts are not an inception completion percentage.



Assessment: this report is ready for stakeholder review; formal acceptance of D1 depends on resolving the outstanding items listed above.

11 Risks and outstanding decisions

11.1 Priority risks

In the table, H means high and M means medium. Likelihood describes how likely the risk is; impact describes how seriously it would affect delivery.

Probability (P) and impact (I) ratings are BITZ's proposed assessment (H/M/L), for validation by project governance. The full register with effects and review notes is in **Annex G**.

R-001	Sandbox and API access delay	H	H	Close checklist (A-015); access or global demonstration system (A-016); escalate if not received by 28 Oct	Rantsala / Isaiah	Open
R-002	Outstanding country documents and SOP applicability	H	H	Obtain A-013/A-017; confirm SOP applicability under A-014; record receipts	Rantsala / Hopolang / UNICEF	Open
R-003	Field and service-routing configuration incomplete	H	H	Service list (A-018); service-to-recipient configuration; mapping approval	Isaiah / Ministry	Open
R-004	Service interruption and duplicate recovery unresolved	M	H	Resolve A-021; adopt stable case reference; test recovery	Nelson / Peter / Isaiah	Open
R-005	ABR verification dependency	M	M	Keep missing ABR non-blocking; A-007, A-019, A-020	Ministry / UNICEF / Isaiah	Open
R-006	Two-way design: disclosure risk or lost updates	M	H	Bounded feedback; reconciliation; webhook role	BITZ / Primero	Open
R-007	Privacy and authority gaps	H	H	A-017, A-022; named approvers	UNICEF / Ministry / Nelson	Open
R-008	Focal-person or support gaps	H	M	A-024 cover; A-002 permanent focal; named support owners	Hopolang	Open
R-009	Milestone slippage	H	H	Governance confirms D1 status; written schedule rebaseline	Project governance	Open
R-010	Contract scope change (two-way, attachments)	H	H	Governance decision (section 9)	UNICEF / governance	Open
R-011	CPIMS+ consent requirement	M	H	Ministry defines consent capture and override; Primero confirms configured checks	Ministry / Isaiah	Open
R-012	Primero country so-live dependency	M	H	Primero reports Lesotho so-live plan	Isaiah / governance	Open
R-013 (new)	Authentication method for the automated connection not yet agreed	M	H	Agree how the connection signs in automatically and renews its access in A-023 working session	Isaiah / Peter / Marlon	Open
R-014 (new)	Stakeholder approval delays; approvers unnamed	H	H	Name approvers; set review window for this draft	UNICEF / Ministry	Open
R-015 (new)	Contract and plan wording on "save-event" trigger conflicts with D-006	M	M	Governance confirms D2/D4 acceptance wording (CD-03)	Project governance	Open

Figure 13 — Risk heat map. Proposed BITZ assessment, for validation. No risk is currently rated Low on either axis.

Probability ↓ / Impact →	High	Medium	Low
High	R-001 access · R-002 country documents · R-003 configuration · R-007 privacy · R-009 milestones · R-010 scope · R-014 approvals	R-008 focal person	—
Medium	R-004 service interruption and duplicates · R-006 two-way design · R-011 consent · R-012 Primero so-live · R-013 authentication	R-005 ABR verification · R-015 save-event wording	—

Probability ↓ / Impact →	High	Medium	Low
Low	—	—	—

Seven risks sit in the High/High cell. Five of them (R-001, R-002, R-003, R-007, R-014) can be cleared only by inputs or decisions from UNICEF, the Ministry or Primero. They are the focus of the actions in section 11.

11.2 Risk review and escalation (Proposed)

1. The risk register is reviewed at each steering review and updated in Annex G and the inventory.
2. A risk rated H/H, or any risk whose mitigation action passes its due date without evidence, is escalated to the UNICEF programme owner and Ministry business owner at the next steering review.
3. A risk is closed only when evidence of its resolution has been recorded.
4. New risks are added with the next free R-number and recorded in the risk register.

11.3 Issues

Review of the contract text identifies further planning issues: confirm the governing contract and execution status (CD-02); reconcile access dates and week references (CD-07/CD-08); name the UNICEF signatory (CD-09); agree administration of support beyond February (CD-10); confirm submission formats (CD-11); and reconcile proposed fault waivers with UNICEF acceptance (CD-12). UNICEF and BITZ should resolve these during review of the recovery timetable. The detailed evidence and treatment are in section 2.8.

Current issues (risks already realised): the D1 target month has passed without approval (R-009); the test system checklist made no progress between Session 2 and Session 3 (R-001); Session 3 actions due 26 October remain outstanding (R-002, R-007).

11.4 Decisions required before detailed design can be approved

1. Minimum child-identification dataset and emergency exceptions (WF-004, DATA-005).
2. Consent or lawful-basis approach and override authority (DATA-012, R-011).
3. Authentication flow for the unattended integration (API-002).
4. Sequence design and service-to-recipient configuration (API-003, RT-003).
5. Duplicate prevention key adoption (OPS-003).
6. Return-update mechanism and permitted fields (RT-007), and contract treatment (R-010).
7. Direct software connection or OpenFN (API-009).

11.5 Decisions required before integration testing can be approved

1. Configured Lesotho environment or explicit acceptance of global demonstration system limitations (D-018, API-007).
2. Service interruption handling (A-021, OPS-001) and retry policy (OPS-002).
3. Approved fictional records for testing (UAT-003).
4. Fault severity and so-live rules (UAT-007).

12 Approval and next steps

Stakeholders are asked to review the business baseline and proposed delivery approach, identify required amendments and name authorised approvers. Approval must state whether unresolved design, access and configuration items are acceptance conditions or outstanding items with an agreed owner, deadline and effect on delivery. Business agreement does not substitute for technical review and testing, a contract variation or formal deliverable acceptance.

12.1 Review and approval record

This review should confirm the governing contract, original milestone record and proposed recovery dates separately. Approval of this report confirms the inception baseline; it does not signify completion of development or testing. UNICEF's five-working-day review period and written acceptance requirements apply as stated in the supplied text [CT1]; the signed contract and named authorities remain to be confirmed.

Approval role	Name	Decision	Date
Ministry business owner	To confirm	Pending	
UNICEF programme / governance owner	To confirm	Pending	
OpenCHS technical authority	To confirm	Pending	
Primero technical authority	To confirm	Pending	
Data protection / security authority	To confirm	Pending	
UNICEF user-testing signatory	To confirm; role specified in [CT1 section 3.3]	Pending	
Ministry user-testing endorsement	Representative and endorsement arrangements to confirm	Pending	
Project coordination (BITZ)	Nelson Adagi	Circulation and review tracking pending	

No approvals have been recorded to date.

12.2 Immediate actions (status as at 28 October 2026)

"Overdue" indicates that the due date has passed and completion has not yet been confirmed to BITZ.

Action	Responsible party	Dependency	Required by	Status
A-013 Share national referral pathways	Rantsala (UNICEF)	—	26 Oct 2026	Overdue — not yet received
A-015 Complete test system checklist; post-meeting call	Rantsala, Isaiah	—	26 Oct 2026	Overdue — outcome not recorded
A-017 Share privacy materials	Lekhanya (UNICEF)	—	26 Oct 2026	Overdue — not yet received
S3-N01 Circulate Session 3 minutes	BITZ	—	26 Oct 2026	Overdue — circulation not confirmed

Action	Responsible party	Dependency	Required by	Status
A-014 Resend case-management SOPs	Hopolang	—	27 Oct 2026	Received; applicability review pending [SOP1]
A-020 Share Berea workshop use cases	Melusi	—	27 Oct 2026	Due 27 October — not yet received
A-024 Name helpline leave cover	Hopolang	—	27 Oct 2026	Due 27 October — not yet confirmed
S3-N02 One-to-one follow-ups	BITZ	—	27 Oct 2026	Due 27 October — outcome not recorded
A-016 Provide test system or global demonstration system access	Isaiah	A-015	28 Oct 2026	Pending
A-018 Confirm CPIMS+ service list	Isaiah	—	28 Oct 2026	Pending
A-021 Agree service interruption handling	Nelson, Peter, Isaiah	—	28 Oct 2026 or during testing	Pending; proposals in section 6
A-022 Data protection desk research	Nelson	A-017	28 Oct 2026	In progress
A-025 Deliver initial draft inception report	Nelson / BITZ	—	28 Oct 2026, close of business	Draft prepared; circulation pending
A-023 Configure API authentication; start test calls	Peter, Marlon	A-016	After A-016	Awaiting dependency
A-019 Invite BITZ to Primero/X-Road sessions	Lekhanya (UNICEF), Isaiah	—	Next session; date not recorded	Pending
A-002 Confirm permanent Child Helpline focal person	Hopolang	—	Not restated	Carried over
A-007 Remaining ABR verification rules	Ministry / relevant authority	A-019, A-020	Not restated	Carried over
P-01 Define two-way update contract	OpenCHS / Isaiah / Ministry	R-010	To agree	Proposed
P-02 Complete the agreed data mapping between the two systems	OpenCHS / Isaiah / Ministry	A-016, A-018	To agree	Proposed
P-03 Confirm UNICEF action owners	UNICEF	—	To agree	Proposed
New — Name approvers for D1	UNICEF, Ministry	—	Before formal review	Proposed
New — Governance decision on scope and schedule	UNICEF, Ministry	—	Before D3 design sign-off	Proposed

Annexes

Annex	Content
A	Detailed requirements matrix
B	API operations and authentication
C	Field mapping and data dictionary status
D	Referral workflow, sequence detail and status vocabulary
E	Decision register
F	Action register
G	Detailed risk register
H	UAT scenario matrix
I	Source documents and references
J	Categorised requirement baseline and status
K	Detailed connection design and security specifications

Annex A — Detailed requirements matrix

Priority values are BITZ proposals (see Annex J). Validation methods: **T** = test (Annex H reference), **R** = review/inspection, **D** = demonstration, **E** = evidence of receipt or approval.

ID	Requirement	Source	Priority	Status	Validation / acceptance
WF-001	Lifecycle as presented in Session 3; aligned to national pathways and SOPs	D-006; [M3 sections 3–4]	Must	Confirmed; Dependency A-013/A-014	R: lifecycle mapped against received SOPs; T: B-01
WF-002	Manual referral by handling agent; save never sends	D-006	Must	Confirmed	T: F-01
WF-003	Optional supervisor escalation; mandatory-escalation rules defined	[M1 section 6]	Should	Pending Confirmation	E: approved rules; T: B-03
WF-004	Emergency minimum-information referral	D-003	Must	Pending Confirmation	E: approved minimum dataset; T: F-05
WF-005	Partial referral and later information	[M2 section 3]	Should	Pending Confirmation	E: approved rule; T: F-13
WF-006	Correction, cancellation, withdrawal and re-referral with history	[TB section 4]	Must	Pending Confirmation	E: approved rules; T: F-19
WF-007	Parallel emergency phone path	[M1 section 8]	Must	Confirmed	R: SOP and training content; T: B-02
WF-008	Case plan excluded	D-012	Must	Confirmed	T: F-16
WF-009	Anonymous reporter; identified child	D-010; D-011	Must	Confirmed	T: F-03, F-04
WF-010	Prank finding sent as follow-up update	[M1 section 7]	Must	Confirmed	T: F-13
RT-001	CPIMS+ service list	A-018	Must	Dependency	E: list received and approved
RT-002	Service selection; Ministry distribution	D-008	Must	Confirmed	T: F-10, F-18
RT-003	Configured recipients, reasons, assignment permissions	D-008	Must	Pending Confirmation	E: configuration record; T: F-10
RT-004	Non-Ministry stakeholders via CPIMS+ accounts	D-001; [M2 section 4]	Could	Pending Confirmation	E: configuration record
RT-005	Technical receipt distinct from human acceptance	D-016	Must	Confirmed	T: F-11
RT-006	Intermediate status vocabulary	[M2 section 5]	Must	Pending Confirmation	E: approved vocabulary (Annex D.3)
RT-007	Follow-up and return-update contract	D-007	Must	Pending Confirmation	E: approved contract; T: F-12, F-13, RC-05
RT-008	Two-way referred-case feedback	D-007	Must	Confirmed	T: F-12
FN-001	Optional ABR field; blank never blocks	D-005	Must	Confirmed	T: F-07
FN-002	ABR structural validation	D-014	Must	Confirmed	T: F-06, F-08, F-09
FN-003	Explicit referral control; double-submit prevention	D-006; [TB section 4]	Must	Confirmed; design Proposed	T: F-01, RC-02
FN-004	Narrative required at referral only	D-013	Must	Confirmed	T: F-02
FN-005	Child identity required at referral	D-011	Must	Confirmed; fields Pending	T: F-04

ID	Requirement	Source	Priority	Status	Validation / acceptance
FN-006	Service selection from maintained list	D-008	Must	Confirmed; Dependency	T: F-10
FN-007	Referral status display	[TB section 4]	Must	Contractual	T: F-11, F-12
FN-008	Separate acknowledgement events displayed	D-016	Must	Confirmed	T: F-11
FN-009	Follow-up updates to linked case	D-012; [M1 section 7]	Must	Confirmed	T: F-13
FN-010	Returned status and interventions on linked case	D-007	Must	Confirmed	T: F-12
FN-011	Case plan never transmitted	D-012	Must	Confirmed	T: F-16
FN-012	Clear validation messages	[TB section 4]	Must	Contractual	T: F-02, F-04, F-08
DATA-001	ABR issuer NICR	D-014	Must	Confirmed	R: provenance retained
DATA-002	ABR optional; caseworker obtains later	D-005	Must	Confirmed	T: F-07
DATA-003	Remaining ABR verification contract	A-007	Should	Pending Confirmation	E: authority response
DATA-004	ABR correction and cross-system update	D-005	Should	Pending Confirmation	T: F-17
DATA-005	Child identity mapping and minimum fields	D-011	Must	Pending Confirmation	E: approved mapping
DATA-006	Age, sex, nationality, location, contact	D-012	Must	Pending Confirmation	E: approved mapping
DATA-007	Protection-concern mapping (multi-select)	[M2 section 11]	Must	Pending Confirmation	E: standard list; T: F-10
DATA-008	Reporter contact and anonymity	D-010	Must	Pending Confirmation	T: F-03
DATA-009	Perpetrator details	D-004; D-012	Must	Pending Confirmation	T: F-14
DATA-010	Attachments controls	D-012	Must	Pending Confirmation	T: F-15
DATA-011	Narrative at referral	D-013	Must	Confirmed	T: F-02
DATA-012	Consent / disclosure representation	Annex B.6	Must	Pending Confirmation	E: Ministry decision; T: S-06
DATA-013	Stable external case reference	OPS-003	Must	Proposed	T: RC-01
API-001	API v2 specification is build basis	D-017	Must	Confirmed	R: version recorded
API-002	Authentication contract	[TR1 section 3]	Must	Pending Confirmation	T: S-01, RC-08
API-003	Operation sequence	Annex B	Must	Pending Confirmation	E: API contract capture; T: F-10
API-004	Errors, timeouts, performance	[TR1 section 8]	Must	Pending Confirmation	T: RC-06
API-005	OpenCHS test/UAT environment	[M1 section 9]	Must	Pending Confirmation	E: access evidence

ID	Requirement	Source	Priority	Status	Validation / acceptance
API-006	Sandbox checklist closed	A-015	Must	Dependency	E: completed checklist
API-007	Sandbox or playground access	D-018; A-016	Must	Dependency	E: authenticated test call
API-008	Production environment and secrets		Must	Pending Confirmation	E: readiness checklist
API-009	Direct connection, not X-Road	D-015	Must	Confirmed	R: architecture
API-010	Upstream API documentation received	D-017	Must	Confirmed	R: reference retained
GOV-004	Deployed release and URLs		Must	Pending Confirmation	E: recorded by Primero
OPS-003	Idempotency key and reconciliation	[TB section 5E]	Must	Pending Confirmation	T: RC-01
SEC-001	Lawful basis and purposes	A-017; A-022	Must	Dependency	E: written confirmation
SEC-002	Reporter anonymity preserved	D-010	Must	Confirmed	T: F-03, S-03
SEC-003	Least-privilege integration account	Annex B.2	Must	Proposed	T: S-02
SEC-004	Secret storage and rotation	[TB section 5B]	Must	Pending Confirmation	R: secret-store configuration
SEC-005	Transport security	Delivery plan	Must	Proposed	T: S-05
SEC-006	Log redaction and access	[TB section 5F]	Must	Pending Confirmation	T: S-03
SEC-007	Synthetic test data	[TR1 section 9]	Must	Proposed	E: approved dataset; T: S-08
SEC-008	Bounded return updates	D-007	Must	Proposed	T: S-04

A.6 Operational and reliability

OPS-001	Durable queue; outage behaviour	[TB section 5D]; A-021	Must	Pending Confirmation	T: RC-03
OPS-002	Retry and escalation policy	Delivery plan	Must	Proposed	T: RC-04
OPS-005	Masking and log access		Must	Pending Confirmation	T: S-03
OPS-006	Monitoring and support owners		Must	Pending Confirmation	E: named owners
OPS-007	Incident and outage procedure	A-021	Must	Pending Confirmation	E: approved procedure
OPS-008	Regression of existing OpenCHS	[TB section 6]	Must	Contractual	T: RG-01

Annex B — API operations and authentication

B.1 Specification and version

The API reference supplied in the Primero technical response [TR1 section 2] describes Primero API v2 (development_v2 branch README). D-017 accepts it as the build basis unless it changes. The development_v2 branch moves over time. For this report the upstream Primero source and API documentation (doc/api/) were reviewed at release **v2.16.0** (commit 622e0d2d6d). **Neither establishes the release or configuration deployed for Lesotho** (GOV-004).

B.2 Operations relevant to the integration (upstream v2.16.0)

Purpose	Operation	Documented behaviour
Obtain or renew token	POST /api/v2/tokens	Username/password, or an existing valid token; token returned in the Authorization header
Invalidate token	DELETE /api/v2/tokens	Expires the token presented
Create case	POST /api/v2/cases	data object; optional client-generated UUID in data.id
Update case	PATCH /api/v2/cases/:id	Merges submitted fields; subform arrays merged and appended
Refer case	POST /api/v2/cases/:id/referrals	Addressed to a receiving user (transitioned_to); accepts service, agency, notes and consent-override fields
Read referral outcomes	GET /api/v2/cases/:id/referrals; GET /api/v2/cases/:id/transitions	Lists referrals and other transitions
Attach file	POST /api/v2/cases/:id/attachments	Base64 file with field name, type and content type; requires permission on that field
Read history	GET /api/v2/cases/:id/record_history	Change history for a case

Case creation and referral are separate operations. A referral grants another user access to the record; a transfer changes ownership through acceptance or rejection. Because an API referral is addressed to a user, service-based routing (D-008) needs a configured mapping from each service to a receiving user or agency (RT-003). The API accepts only fields the integration account's role permits it to set, so that role must cover the approved mapping and no more (SEC-003).

B.3 Case creation and duplicate protection

When a client-generated UUID is supplied, case creation returns 204 No Content on success and 409 Conflict if a case with that identifier exists. Other documented errors are 403 Forbidden and 422 Unprocessable Entity (naming the invalid field). Errors share a code / resource / message structure.

Proposed idempotency approach: OpenCHS generates and stores the UUID before the first attempt and reuses it on every retry; a repeat after a lost response yields 409 rather than a second case. Trade-off: 204 returns no case content, so the case display identifier must be read separately. Adoption requires confirmation that the Lesotho deployment behaves as documented and agreement on 409 reconciliation (OPS-003).

B.4 Authentication

The upstream documentation describes a bearer JWT returned in the response Authorization header, a 60-minute expiry, issuance by username/password, and reissue using a valid existing token; reissue does not invalidate other unexpired tokens. Isaiah prefers SSO where an identity provider exists, with username/password

as the alternative, and confirms that Primero will set up credentials once implementation is ready. The unattended backend flow, expiry recovery, permissions and secret rotation remain **Pending Confirmation** (API-002, R-013). Choosing SSO for users does not by itself settle integration authentication.

Isaiah confirms network/security controls apply without naming them. Obtain the actual TLS/certificate, VPN, IP allowlisting and firewall prerequisites. Isaiah reports no API rate limits and proposes joint performance management; concurrency, payload limits, timeouts and performance acceptance remain to be agreed (API-004).

B.5 Return-update options

Primero supports outbound webhooks that send a record's data to an external URL on create or update, authenticated by API key or basic authentication. Fields sent are limited to those permitted to a configured role; each delivery is logged in Primero's audit log; webhooks are enabled per module by configuration. In v2.16.0 the webhook job logs errors and **does not retry failed deliveries**.

Option	Description	Strengths	Constraints
1. Webhook + reconciliation	Primero posts updates to an authenticated OpenCHS endpoint; OpenCHS periodically reconciles via referral/transition endpoints	Near-real-time; role-based field minimisation	Needs inbound endpoint and network path; missed deliveries need reconciliation; Lesotho enablement and whether referral status changes trigger a webhook unconfirmed
2. Polling	OpenCHS periodically reads referral and transition status for open referred cases	Outbound-only network path; simpler security	Update lag; API load; intervention details may need case reads

Decision: **Pending Confirmation** (RT-007) with Primero and the Ministry.

B.6 Consent check at referral

In the upstream child-protection module, a user referral checks that the case records consent for disclosure to other organisations and consent for services, unless consent is overridden. No consent capture for this purpose was discussed in the sessions, and the corresponding OpenCHS fields have not been verified (DATA-012, R-011).

B.7 API contract still required for Lesotho

- Sandbox/UAT and production base URLs and the deployed release.
- Integration account, authentication flow and permissions.
- Token expiry and renewal handling for the unattended backend.
- Operation sequence for create, referral, follow-up and attachments.
- Configured request and response examples for the Lesotho forms.
- Confirmation of client-generated identifier support and conflict handling.
- Error classes, timeouts and retry/manual-correction rules.
- Service-to-recipient configuration and lookup/form endpoints if needed.
- Return-update mechanism, payload and delivery guarantees.
- TLS, certificate, VPN, IP allowlisting or firewall requirements.

B.8 Configuration dependencies

Primero configures the Lesotho demo from SOPs, paper referral forms, approved child-journey use cases and referral pathways; Isaiah expects field definitions and lookup lists after setup. Isaiah reports no existing interface specification, so the teams must jointly capture one (template 04_templates/06_api_contract_capture.docx).

Annex C — Field mapping and data dictionary status

C.1 Confirmed information set (D-012) and mapping status

The field set is a **business payload baseline**, not a deployed API schema.

Information	Business requirement	Status	Technical mapping still needed
Child name and location	Identifying details required to trace the child (D-011)	Confirmed	Name parts, minimum identity, location hierarchy, exceptions
Age, sex, nationality, child contact	In agreed bio-data set	Confirmed	Types, codes, estimated age, unavailable contact
Case category	Maps to CPIMS+ protection concerns (multi-select)	Confirmed	Approved concern list, multi-value codes
Case narrative	Mandatory at referral, not first save (D-013)	Confirmed	Target field, length, permitted sensitive content
Reporter contact	Included when available; anonymous reporter allowed (D-010)	Confirmed	Anonymous flag/value, optional contact, privacy
Perpetrator details	Included	Confirmed	Fields, unknowns, access, conditional validation
Attachments	Included	Confirmed	File types, size, scanning, permissions, retention
Follow-up updates	Included; linked to the original case	Confirmed	Events, ownership, timestamps, permissions
ABR ID	Optional; NICR structural validation (D-005, D-014)	Confirmed	Target field, blank representation, correction
Selected service	Agent selects from CPIMS+ service list (D-008)	Confirmed; list is a Dependency	Service codes; receiving user or agency per service
Consent and disclosure	Not discussed in sessions	Pending Confirmation	Whether CPIMS+ consent fields must be set (B.6)
CPIMS+ feedback	Status and interventions returned (D-007)	Confirmed	Approved values, minimisation, delivery, reconciliation
Helpline case plan	Not transmitted	Excluded (Confirmed)	Exclude from payload
External case reference	Technical linkage	Proposed	Client-generated UUID or other key; returned case ID

C.2 ABR identifier rules (D-005, D-014)

Attribute	Rule	Status
Meaning	Birth registration number issued at birth; persists through national ID, passport and school records [M1 section 5]	Confirmed
Issuing authority	NICR, Ministry of Home Affairs	Confirmed (reported by Hopolang after contact with Home Affairs)

Attribute	Rule	Status
Length	12 digits	Confirmed
Prefix	First digit 0 or 1	Confirmed
Sex position	Seventh digit: 1 male, 2 female	Confirmed
Requiredness	Optional; referral proceeds when unavailable	Confirmed
Missing identifier	Receiving caseworker obtains it during follow-up	Confirmed
Storage	Text, preserving leading zeroes	Proposed
Sex mismatch handling	Warn, block or accept	Pending Confirmation
Checksum / uniqueness / live verification	Not supplied; A-007 open; Home Affairs route via A-019 sessions	Pending Confirmation

Figure 14 — ABR ID structure (D-014; [M3 section 9]; [RC1] item 6). The source defines only positions 1 and 7. The meaning of the other positions is not recorded.

Rule

1: 0 or 1

2: digit

3: digit

4: digit

5: digit

6: digit

7: 1 = male, 2 = female

8: digit

9: digit

10: digit

11: digit

12: digit

Validation outcomes: a blank value is accepted (D-005). A value that is not 12 digits, starts with 2–9, or has a seventh digit other than 1 or 2 is rejected with a message (F-08). A seventh digit that conflicts with the recorded sex follows the rule still to be agreed (F-09).

Structural checks do not prove that an identifier exists or belongs to the child.

C.3 Mapping attributes required per field

Source and target field; direction; datatype; requiredness (including conditional); validation; transformation; lookup values; sensitivity classification; approving owner. Isaiah supplies configured definitions and lookups; Ministry and OpenCHS stakeholders approve the business mapping (Proposed approval route). Template: 04_templates/04_field_mapping_worksheet.docx.

C.4 Candidate field names

Earlier developer examples (abr_id, openchs_reference_id, name_first, name_last, protection_concerns, location_current, referral_notes) are **candidate names, not the confirmed Lesotho schema**. Date-of-birth, risk-level, module and referral-source examples require explicit mapping approval; their presence in earlier material does not expand the agreed field set. Primero will produce one standardised protection-concern list aligned

across both systems [M2 section 11]. The OpenCHS “Services Offered” value sent on referral is a label such as “CPIMS+” or “Child Protection Service”.

C.5 Data dictionary status

Item	Status
OpenCHS case/contact data object	Shared by Nelson in Session 2
CPIMS+ required field list	Requested in Session 2; not received
Configured Lesotho field definitions and lookups	Dependency — after sandbox setup
Unified national referral form	Recorded as shared (A-004); not available
Standardised protection-concern list	Dependency — Primero
CPIMS+ service list	Dependency — A-018

Annex D — Referral workflow, sequence detail and status vocabulary

D.1 Agreed referral lifecycle

1. The agent receives the call, records the contact and saves the helpline case. The narrative need not be complete at first save (D-013).
2. The handling agent decides to refer and selects an approved CPIMS+ service (D-006, D-008), optionally escalating to a supervisor first [M1 section 6].
3. OpenCHS validates the narrative and required child-identifying information, allowing an anonymous reporter and absent ABR ID (D-010, D-011, D-005).
4. The integration submits the agreed data and records a system acknowledgement or a failure requiring recovery (D-016).
5. The receiving party accepts or rejects the referral through the CPIMS+ workflow. System receipt does not prove human acceptance (D-016).
6. Ministry users distribute and assign internally; the helpline does not select a caseworker (D-008).
7. OpenCHS sends follow-up information against the linked case; CPIMS+ returns approved status and intervention updates (D-007, D-012).
8. Closure, correction, cancellation and re-referral retain linkage and an auditable history; detailed rules pending (WF-006).

Urgent cases are phoned to police or first responders in parallel (WF-007).

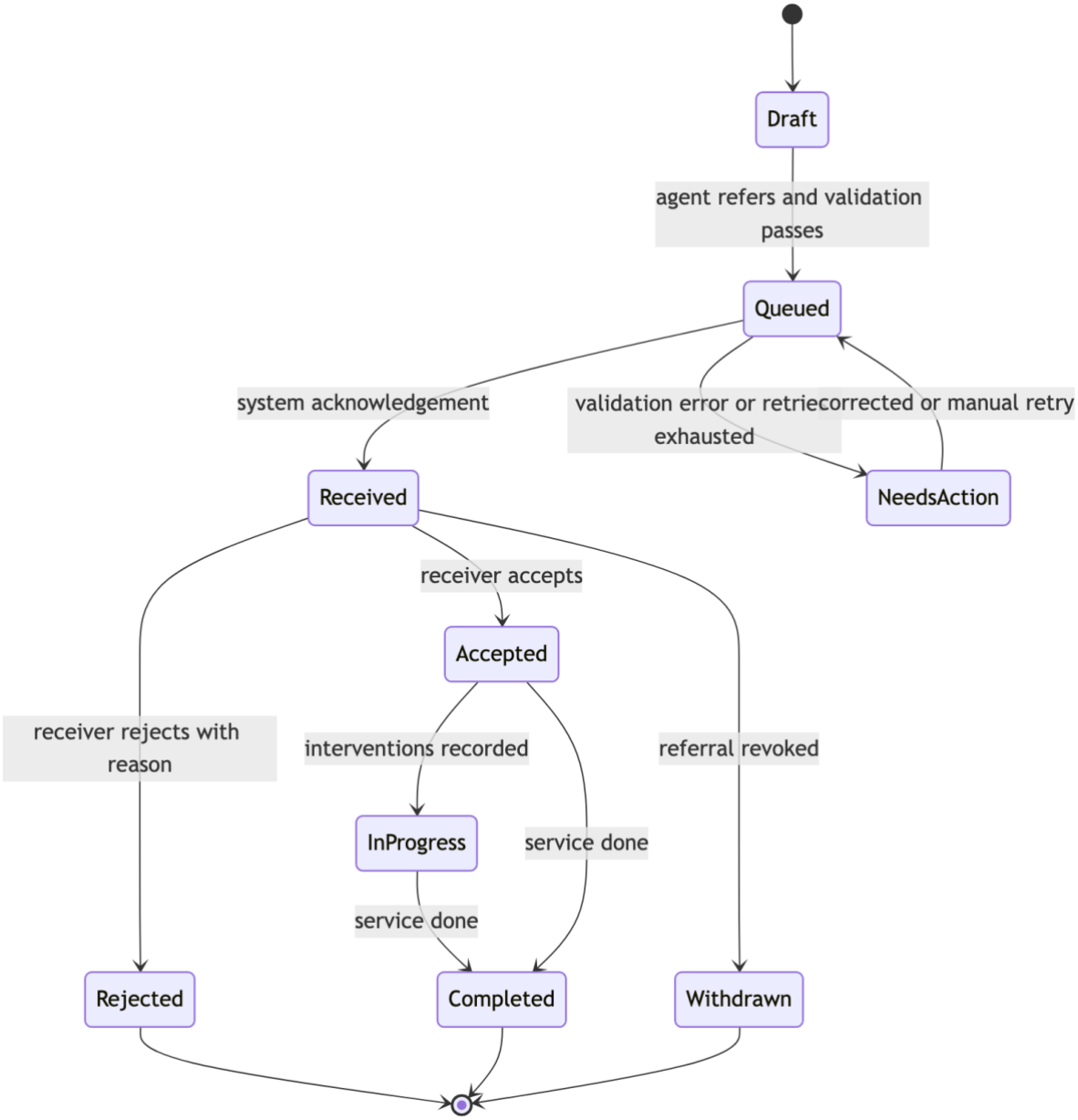
D.2 Native CPIMS+ referral mechanism (Session 2 demonstration)

A user opens the Services panel, chooses an Internal or External referral and a service (for example case management or medical service), and records the reason and expected outcome. The recipient sees the referral as pending and must accept it (which opens the case file) or reject it with a reason. Assignment to a caseworker by a district manager is a separate, later step. The helpline needs visibility of acceptance and later service completion or closure, not of internal assignment [M2 section 4].

D.3 Status vocabulary

Upstream Primero transition statuses (v2.16.0): in_progress, accepted, rejected, done, revoked, with successful/not-successful outcome and whether the service was implemented. The helpline-facing vocabulary should be derived from these rather than invented separately (RT-006).

Figure 15 — Proposed helpline referral states (Pending Confirmation).



Proposed helpline state	Upstream basis	Meaning
Draft	—	Case saved; not referred
Queued	—	Referral accepted by OpenCHS for delivery
Needs action	—	Validation error or delivery failure needing operator/administrator action
Received	API success	System acknowledgement (D-016)
Accepted	accepted	Human acceptance (D-016)
Rejected	rejected	Human rejection with reason
In progress	Intervention updates	Case being worked; interventions returned
Completed	done	Service completed, with outcome
Withdrawn	revoked	Referral revoked

D.4 Reliability design (Proposed)

Durable referral records; stable correlation reference; auditable attempts. Retries reuse the client UUID; uncertain timeouts reconciled; errors classified as retryable or needing correction; proposed back-off at 1 minute, 5 minutes, 30 minutes and 2 hours before marking failed and alerting an administrator (delivery plan; Proposal). Equivalent controls apply to repeated or out-of-order return updates; conflicting updates need explicit ownership rather than blind overwrites. An OpenCHS change must not automatically close or overwrite a CPIMS+ case without an agreed update contract.

Annex E — Decision register

Identifiers continue the Session 1–2 register; where Session 3 updated an earlier decision it kept the earlier ID. Earlier v0.4 report-local numbering is not equivalent.

E.1 Current decisions

ID	Decision	Source	Status	Supersedes
D-001	Support referrals to CPIMS+/MGYSD and other approved stakeholders	Session 2	Confirmed (in principle)	—
D-002	SOP covers the broader referral ecosystem and role-holders	Session 2	Confirmed (in principle)	—
D-003	Emergency referrals need not capture every detail first	Session 2	Confirmed direction; minimum dataset Pending	—
D-004	Perpetrator information is important	Session 2	Confirmed; included by D-012	—
D-005	ABR optional; receiving caseworker obtains it if missing	Session 3	Confirmed	Session 2 D-005 (“subject to approved Ministry process”)
D-006	Manual handling-agent referral; process as presented	Session 3	Confirmed	Session 2 D-006 (“direction discussed”)
D-007	Two-way integration for referred-case updates	Session 3	Confirmed	Session 2 D-007 (closed-loop “in principle”); Session 1 decision 5 one-way scope
D-008	Ministry distribution; helpline selects CPIMS+ service	Session 3	Confirmed; service list to confirm	Session 2 D-008; Session 1 decision 6 (direct to district supervisor)
D-009	Ministry-led UAT with recorded participant roles	Session 3	Confirmed	Session 2 D-009 (“agreed direction”)
D-010	Anonymous reporter allowed, recorded as such in CPIMS+	Session 3	Confirmed	Session 2 position that anonymous reports cannot be referred
D-011	Child identifying details mandatory for referral	Session 3	Confirmed	—
D-012	Field set as presented; case plan not sent	Session 3	Confirmed	Session 3 facilitation-deck proposal to defer attachments
D-013	Narrative mandatory at referral, not first save	Session 3	Confirmed	—
D-014	NICR ABR format for initial validation	Session 3	Confirmed	Session 1 “validation deferred”

ID	Decision	Source	Status	Supersedes
D-015	Direct API or OpenFN; X-Road only for Home Affairs checks	Session 3	Confirmed; API vs OpenFN choice Pending (Ministry)	—
D-016	Two acknowledgements: system and human	Session 3	Confirmed	—
D-017	Primero API documentation is the build basis unless it changes	Session 3	Confirmed	—
D-018	Global playground if Lesotho sandbox not ready by draft report	Session 3	Confirmed	—

“Confirmed” means agreed in the minutes; approval of this report remains pending.

E.2 Session 1 numbered decisions

Session 1 #	Decision	Current position
1	Manual trigger by handling agent	Retained (D-006)
2	Perpetrator details included	Retained (D-004, D-012)
3	Payload: bio-data, category, narrative, reporter contact, perpetrator	Extended by D-012 (attachments, follow-ups, ABR)
4	ABR field optional; validation deferred	Updated by D-005, D-014
5	Post-referral updates pushed to CPIMS+; return not committed	Retained for outbound; return now in scope (D-007)
6	Helpline refers directly to district supervisor	Superseded by D-008
7	Emergencies by direct phone in parallel	Retained (WF-007)
8	OpenCHS integrates directly with CPIMS+; X-Road separate	Retained (D-015)

Annex F — Action register

Status at 27 October 2026. “Completed” is used only where the minutes record completion.

F.1 Session 3 actions

ID	Action	Owner	Recorded due	Status
A-013	Share national referral pathways	Rantsala	26 Oct 2026	Overdue — not yet received
A-014	Resend case-management SOPs incl. referral section	Hopolang Mosena	27 Oct 2026	Received [SOP1]; Ministry to confirm current applicability; BITZ to complete mapping review
A-015	Complete sandbox checklist; post-meeting call	Rantsala / Isaiah	26 Oct 2026	Overdue — outcome not recorded
A-016	Provide Lesotho sandbox or global playground access	Isaiah	28 Oct 2026	Pending
A-017	Share Ministry data-protection work, policy and legislation	Lekhanya (UNICEF), as recorded	26 Oct 2026	Overdue — not yet received; owner to confirm
A-018	Confirm CPIMS+ service list	Isaiah	28 Oct 2026	Pending
A-019	Invite BITZ to Primero/X-Road sessions	Lekhanya (UNICEF) / Isaiah	Next session	Pending
A-020	Share Berea X-Road workshop use cases	Melusi	27 Oct 2026	Due 27 October — not yet received
A-021	Agree handling of referrals during outages	Nelson / Peter / Isaiah	28 Oct 2026 or during testing	Pending
A-022	Data protection desk research	Nelson	28 Oct 2026	In progress
A-023	Configure API authentication; begin test calls	Peter / Marlon	After A-016	Awaiting dependency
A-024	Name helpline coordination leave cover	Hopolang	27 Oct 2026	Due 27 October — not yet confirmed
A-025	Deliver initial draft inception report	Nelson / BITZ	28 Oct 2026, COB	Draft prepared; circulation pending
S3-N01	Circulate Session 3 minutes	BITZ	26 Oct 2026	Circulation not confirmed
S3-N02	One-to-one follow-ups (outage, service list, cover)	BITZ	26–27 Oct 2026	Outcome not recorded
P-01	Define two-way update contract	OpenCHS / Isaiah / Ministry	To agree	Proposed
P-02	Complete configured field mapping	OpenCHS / Isaiah / Ministry	To agree	Proposed
P-03	Confirm UNICEF action owners	UNICEF	To agree	Proposed

F.2 Earlier actions (Sessions 1–2)

ID	Action	Owner	Status as last recorded	Current treatment
A-001	Provide participant list	Hopolang	Completed (Session 2 minutes)	Closed
A-002	Name Child Helpline focal person	Hopolang	Pending	Carried forward by Session 3
A-003	Share referral workflow/process	Tiisetso	Completed (Session 2 minutes)	Closed; document not available
A-004	Share referral form	Isaiah	Completed (Session 2 minutes)	Closed; form not available
A-005	Review referral form for field requirements	Project team	Pending	Not carried forward; field set settled by D-012
A-006	Identify ABR integration support	Hopolang	Pending	Not carried forward; addressed by D-014, A-019
A-007	Confirm ABR format, ownership, rules	Ministry / relevant authority	Pending	Carried forward for residual verification rules
A-008	Agree CPIMS+ recipients and routing	Ministry / CPIMS+	Pending	Not carried forward; D-008, A-018, RT-003
A-009	Define closed-loop feedback statuses and mechanism	Ministry / CPIMS+ / CHL	Pending	Not carried forward; D-007, P-01, RT-006/007
A-010	Set up sandbox and API access	UNICEF / technical teams	Pending	Not carried forward; A-015, A-016
A-011	Confirm UAT participants, data, scenarios, sign-off	Ministry / CHL / technical	Pending	Not carried forward; D-009, UAT-002/003
A-012	Confirm UNICEF participation or approval	UNICEF / governance	Pending	Participation addressed in Session 3; approval route GOV-003

Annex G — Detailed risk register

Risk ID	Description	Effect	P	I	Mitigation	Owner	Trigger for escalation	Status
R-001	Sandbox / API access delay	Blocks configured mapping and integration tests	H	H	A-015 checklist; A-016 access or playground; escalation template 14	Rantsala / Isaiah	No access by 28 Oct	Open
R-002	Outstanding country documents and SOP applicability	Requirements cannot be checked against approved practice	H	H	Obtain A-013/A-017; confirm SOP applicability; reconcile Form 4 with the unified referral form; retain receipts	Rantsala / Hopolang / UNICEF	Any item overdue at next review	Open
R-003	Field and service configuration incomplete	Invalid payloads or misrouting	H	H	A-018 service list; service-to-recipient configuration; approved mapping	Isaiah / Ministry	No list by 28 Oct	Open
R-004	Outage and duplicate recovery unresolved	Lost, repeated or uncertain referrals	M	H	Resolve A-021; adopt client UUID; recovery tests RC-01–RC-04	Nelson / Peter / Isaiah	Not agreed before design sign-off	Open
R-005	ABR verification dependency	Format-valid IDs may remain unverified	M	M	Keep missing ABR non-blocking; A-007, A-019, A-020	Ministry / UNICEF / Isaiah	Home Affairs route demands build scope	Open
R-006	Two-way design	Disclosure risk or lost updates	M	H	Bounded feedback; reconciliation; webhook role	BITZ / Primero	Mechanism undecided at design review	Open
R-007	Privacy and authority gaps	Unauthorised sharing or unclear approvals	H	H	A-017, A-022; named approvers	UNICEF / Ministry / Nelson	Materials not received by next review	Open
R-008	Focal-person or support gaps	Delayed review, UAT and incident resolution	H	M	A-024 cover; A-002 focal; named support owners	Hopolang	No cover named by 27 Oct	Open
R-009	Milestone slippage	D1 overdue; D2/D3 windows at risk	H	H	Governance confirms D1 status; written rebaseline	Project governance	Already realised (issue)	Open
R-010	Contract scope change	Two-way updates and attachments exceed outbound D3	H	H	Governance decision (section 9)	UNICEF / governance	No decision before D3 design sign-off	Open
R-011	CPIMS+ consent requirement	Referrals blocked or made without agreed basis	M	H	Ministry defines consent capture and override; Isaiah confirms configured checks	Ministry / Isaiah	Consent check active in configured environment without decision	Open
R-012	Primero go-live dependency	OpenCHS delivery pushed outside window	M	H	Primero reports Lesotho go-live plan	Isaiah / governance	No plan by next steering review	Open
R-013	Unattended authentication flow undecided	Integration cannot run unattended or credentials handled insecurely	M	H	Agree service-account flow, renewal and storage	Isaiah / Peter / Marlon	Not agreed when access arrives	Open (new, proposed)
R-014	Stakeholder approval delays	D1 cannot be accepted; payment and schedule affected	H	H	Name approvers; agreed review window	UNICEF / Ministry	No approvers named by draft circulation	Open (new, proposed)
R-015	“Save-event” wording in plan conflicts with D-006	Misbuild or acceptance dispute for D2/D4	M	M	Confirm acceptance wording (CD-03)	Project governance	Not clarified before D2 demonstration	Open (new, proposed)

P and I: proposed BITZ assessment, **not formally assessed**. All risks remain open until closure is evidenced.

Annex H — UAT scenario matrix

Planned scenarios, **not executed tests**. Each needs expected result, tester, environment, evidence and defect outcome at execution. Priority values are proposed.

H.1 Business process (B)

ID	Scenario	Basis	Expected result	Priority
B-01	End-to-end referral following SOP pathway	WF-001; D-006	Steps D.1 1–7 complete with correct states	Must
B-02	Emergency case: phone call plus system referral	WF-007; D-003	Phone path unaffected; referral recorded and tracked	Must
B-03	Agent escalates to supervisor before referral	WF-003	Escalation recorded; referral by agent after review	Should

H.2 Functional (F)

ID	Scenario	Basis	Expected result	Priority
F-01	Save case without referring	D-006	Nothing sent to CPIMS+	Must
F-02	Save without narrative; attempt referral without narrative	D-013	Save succeeds; referral blocked with message	Must
F-03	Anonymous reporter, identified child	D-010; D-011	Referral succeeds; CPIMS+ shows reporter anonymous; no reporter identity sent	Must
F-04	Missing child identifying details	D-011	Referral blocked with message (unless approved emergency exception)	Must
F-05	Emergency minimum-information referral	D-003; WF-004	Behaves per approved minimum dataset	Must (rule pending)
F-06	Valid ABR ID	D-014	Accepted and transmitted	Must
F-07	ABR absent	D-005	Referral proceeds	Must
F-08	ABR invalid: 11 or 13 digits; prefix 2–9; seventh digit not 1 or 2; non-numeric	D-014	Clear validation message; leading zero preserved for valid IDs	Must
F-09	ABR seventh digit inconsistent with recorded sex	D-014	Behaves per agreed rule	Should (rule pending)
F-10	Service selection routes to configured recipient	D-008	Referral appears for the configured receiving user/agency	Must
F-11	System acknowledgement, then accept; separately reject with reason	D-016	Distinct Received, Accepted / Rejected states and reason shown	Must
F-12	Status and intervention updates returned	D-007	Linked OpenCHS case shows approved fields only	Must
F-13	OpenCHS follow-up, including prank re-flag	D-012; [M1 section 7]	Update applied to linked CPIMS+ case, no new case	Must

ID	Scenario	Basis	Expected result	Priority
F-14	Perpetrator details	D-004; D-012	Transmitted to agreed fields; unknowns handled	Must
F-15	Attachment within and outside agreed limits	D-012	Allowed files transmitted; disallowed rejected with message	Must
F-16	Case plan present in OpenCHS	D-012	Case plan not in payload	Must
F-17	ABR added or corrected later	DATA-004	Behaves per agreed update contract	Should
F-18	Ministry distribution after acceptance	D-008	Helpline sees acceptance and progress, not internal assignment	Must
F-19	Correction, cancellation, re-referral	WF-006	Behaves per approved rules; history preserved	Must (rules pending)

H.3 Security and privacy (S)

ID	Scenario	Basis	Expected result	Priority
S-01	Token issue, expiry after 60 minutes, renewal	API-002	Uninterrupted delivery; no credential in logs	Must
S-02	Integration account attempts unapproved field	SEC-003	Field not set	Must
S-03	Log inspection	SEC-006	No tokens or full narratives; reporter anonymity intact	Must
S-04	Return-update channel authentication and field limits	SEC-008; RT-007	Unauthenticated calls refused; only approved fields stored	Must
S-05	Transport security	SEC-005	TLS enforced; invalid certificate rejected outside local	Must
S-06	Consent fields and override	DATA-012	Behaves per Ministry decision	Must
S-07	OpenCHS role access to returned data	SEC-008	Only authorised roles see returned updates	Must
S-08	Test data check	SEC-007	Only approved synthetic data in test environments	Must

H.4 Recovery and reliability (RC)

ID	Scenario	Basis	Expected result	Priority
RC-01	Lost response after case creation; retry with same UUID	OPS-003	409 reconciled; exactly one CPIMS+ case	Must
RC-02	Agent double-clicks referral	FN-003	One referral only	Must
RC-03	Primero unavailable; referrals queued; service restored	OPS-001	All queued referrals delivered or flagged; none lost	Must
RC-04	Retries exhausted	OPS-002; AUD-001/002	Failed state, administrator alert, controlled manual retry works	Must
RC-05	Duplicate and out-of-order return updates	RT-007	Duplicates ignored; stale updates do not overwrite newer state	Must

ID	Scenario	Basis	Expected result	Priority
RC-06	422 validation error from Primero	API-004	Needs-action state with field named; no blind retry	Must
RC-07	Missed webhook delivery (if Option 1)	B.5	Reconciliation recovers the update	Must (if Option 1)
RC-08	Token expires while queue is draining	API-002	Re-authentication; delivery continues	Must

H.5 Audit and regression

ID	Scenario	Basis	Expected result	Priority
AU-01	Audit record completeness	OPS-004	Every transmission and return update has all approved fields	Must
RG-01	Regression of existing OpenCHS functions	OPS-008	No Critical/High regression	Must

H.6 Coverage of Session 3 decisions

Decision	Scenarios
D-005	F-07, F-17
D-006	F-01, B-01, RC-02
D-007	F-12, RC-05, S-04, RC-07
D-008	F-10, F-18
D-009	UAT organisation (section 10.2)
D-010	F-03, S-03
D-011	F-03, F-04
D-012	F-13, F-14, F-15, F-16
D-013	F-02
D-014	F-06, F-08, F-09
D-015	Architecture review (section 6)
D-016	F-11
D-017	S-01, RC-01, RC-06 (built to documented API)
D-018	Environment record notes playground vs Lesotho configuration for each test

Annex I — Source documents and references

I.1 Source register

The bracketed codes are used for citations throughout the report. Where a source has numbered sections, citations take the form [M3 section 11].

Ref.	Source	Location	Date	Use
[M3]	OpenCHS & Primero CPIMS+ Integration — Session 3 Minutes	03_meetings/session_3/OpenCHS & Primero CPIMS+ Integration — Session 3 Minutes.pdf	26 Oct 2026	Governing baseline: D-005–D-018; A-013–A-025
[AT3]	Session 3 Action Tracker	03_meetings/session_3/SESSION_3_ACTION_TRACKER_2026-10-26.xlsx	26 Oct 2026	Actions, S3-N, P-01–P-03, contacts
[M2]	OpenCHS & Primero CPIMS+ Integration — Session 2 Minutes (Technical Architecture)	03_meetings/session_2/SESSION_2_MINUTES.pdf	9 Oct 2026	D-001–D-004; CPIMS+ demonstration; earlier actions
[M1]	OpenCHS & Primero CPIMS+ Integration — Session 1 Minutes (Business & Workflow)	03_meetings/session_1/SESSION_1_MINUTES.pdf	8 Oct 2026	Workflow, attendees, original decisions
[TR1]	Primero Technical Response to the BITZ inception questionnaire (request dated 2 Oct 2026); section numbers follow the questionnaire	06_correspondence/ISAIAH_RESPONSE_SOURCE_2026-10-26.md (transcription)	Recorded 26 Oct 2026; reply date not supplied	Environment, API, authentication, error handling, test data, UAT
[RQ1]	BITZ circulated requirements and ownership checklist (email)	06_correspondence/NELSON_LAST_CIRCULATED_REQUIREMENTS_EMAIL_SOURCE_2026-10-26.md	Sent date not supplied	Outstanding checklist; 14 Oct call reference
[CT1]	Owner-supplied TICAD-2026-001 contract text; planning extract	Contract planning extract	Supplied 28 Oct 2026; signed status unverified	Sections 3–7 and 9–10: schedule, dependencies, responsibilities, reporting, approvals and support
[RC1]	CHL Integration with Primero Requirement Checklist (annotated); item numbers follow the checklist	03_meetings/CHL intergration with Primero Requirement Checklist.docx	Author and date not recorded	NICR format (item 6); UAT participants and sign-off (item 10); data protection reference (item 11)
[TB]	Technical Scope / TOR Brief — Lesotho Child Helpline 116 / OpenCHS, Amendment 3	07_source_documents/Lesotho_OpenCHS_Technical_Scope_Inception_Brief.pdf	24 Sep 2026	Contract scope; workstreams; inception questions
[DP]	Contract-Aligned Delivery Plan v0.3	02_project_management/CONTRACT_ALIGNED_DELIVERY_PLAN.md	14 Oct 2026; updated 26 Oct 2026	Milestones; payment gates

Ref.	Source	Location	Date	Use
[CI]	Inception Completion Inventory v0.3	01_inception/INCEPTION_COMPLETION_INVENTORY.md	26 Oct 2026	Requirement IDs and status
[R06]	D1 Inception Report v0.6	01_inception/DRAFT_INCEPTION_REPORT_D1_OPENCHS_PRIMERO.md	27 Oct 2026	Predecessor
[PA]	Primero source and API v2 documentation, release v2.16.0	Local repository, commit 622e0d2d6d; doc/api/; public reference https://github.com/primeroIMS/primero/blob/development_v2/app/controllers/api/README.md	Commit 19 Oct 2026; public reference reviewed 26 Oct 2026	Annex B
—	Technical reference material	05_technical_reference/	Various	Prior design context only
—	Capture templates	04_templates/	26 Oct 2026	Working forms; pre-Session 3 content must be reconciled

I.2 Sources referenced but not yet available

Source	Status
Signed governing contract and general terms	TICAD-2026-001 text supplied [CT1]; signed original and Annex A not yet provided. Relationship to 43427866 Amendment 3 remains unresolved
Minutes of the 24 Sep 2026 inception meeting	Not available
Minutes or outcome of the 14 Oct 2026 technical call	Not available
National referral pathways (A-013)	Not received as at 27 Oct
Case-management SOPs (A-014)	Received and reviewed for report impact on 28 Oct [SOP1]; September 2019 edition; current applicability pending Ministry confirmation
Unified national referral form (A-004)	Recorded as shared; not available
Privacy materials and legislation (A-017)	Not received as at 27 Oct
Berea X-Road workshop use cases (A-020)	Not received as at 27 Oct
Sandbox readiness checklist (A-015)	Not available
CPIMS+ service list (A-018)	Not received as at 27 Oct
Configured Lesotho field definitions and lookups	Dependent on sandbox setup
Original Isaiah email	Transcription only; sent date not supplied

I.3 Acronyms retained from sources

ABR ID; API; CHL; CPIMS / CPIMS+; D1–D4; GBV; JWT; MGYSD; NICR; OpenCHS; OpenFN; SOP; SSO; TLS; TOR; UAT; UNICEF; VPN; X-Road. Expansions are given only where the sources state them (NICR: issued by the Ministry of Home Affairs).

Annex J — Categorised requirement baseline and status

Contract update: earlier checklist approval roles and proposed technical options must be read with [CT1] and sections 2.8, 9 and 10. UNICEF executes contractual user-testing sign-off; named signatory remains pending. The official ABR check-digit specification remains required despite optional ABR entry.

These detailed records preserve the 68-item baseline formerly in section 7 of v0.7. Annex A provides the wider requirements traceability and validation matrix. Neither table is evidence of implementation or approval.

J.1 Business and workflow requirements

ID	Requirement	Source	Priority (proposed)	Status
WF-001	Referral follows the lifecycle presented in Session 3, aligned to existing national pathways and SOPs; BITZ does not draft a new national pathway	D-006; [M3 sections 3–4]	Must	Confirmed; SOP received, applicability pending; separate pathways outstanding (A-013, A-014)
WF-002	Referral is initiated manually by the handling agent; saving a case does not send a referral	D-006; [M1 section 6]	Must	Confirmed
WF-003	Agent may escalate to a supervisor before referring; rules for mandatory escalation to be defined	[M1 section 6]	Should	Pending Confirmation
WF-004	Emergency referrals need not capture every detail first; minimum dataset to be defined	D-003	Must	Pending Confirmation
WF-005	Partial referrals and later information handled with a defined status and completion rule	[M2 section 3]	Should	Pending Confirmation
WF-006	Correction, cancellation, withdrawal and re-referral rules with preserved history	[TB section 4] Q6–Q7	Must	Pending Confirmation
WF-007	Urgent cases continue to be phoned to police or first responders in parallel	[M1 section 8]	Must	Confirmed
WF-009	Reporter may be anonymous; child's identifying details are required for referral	D-010; D-011	Must	Confirmed
RT-002	Agent selects an approved CPIMS+ service; Ministry users distribute; helpline does not select a caseworker	D-008	Must	Confirmed; service list is a Dependency (A-018)
RT-003	Configured receiving users or agencies, accept/reject reasons and assignment permissions	D-008; [M2 section 4]	Must	Pending Confirmation
RT-004	Routing to non-Ministry stakeholders (police, courts) via CPIMS+ accounts	D-001; [M2 section 4]	Could	Pending Confirmation
WF-010 *(new)*	A case later found to be a prank is reported to CPIMS+ as a follow-up update	[M1 section 7]	Must	Confirmed (Session 1, not superseded)

J.2 Functional requirements

ID	Requirement	Source	Priority (proposed)	Status
FN-001	Optional ABR ID field on the OpenCHS intake form; blank value never blocks saving or referral	D-005; [TB section 4]	Must	Confirmed
FN-002	ABR structural validation: 12 digits; first digit 0 or 1; seventh digit 1 (male) or 2 (female); clear message when invalid	D-014	Must	Confirmed (handling of sex mismatch Pending Confirmation)
FN-003	Explicit referral control available to the handling agent; prevents accidental or double submission	D-006; [TB section 4]	Must	Confirmed (contract); design Proposed
FN-004	Referral blocked until case narrative is present; case can be saved without it	D-013	Must	Confirmed
FN-005	Referral blocked until required child identifying details are present, subject to approved emergency exceptions	D-011; WF-004	Must	Confirmed; minimum fields Pending Confirmation
FN-006	Agent selects one approved CPIMS+ service from a maintained list	D-008	Must	Confirmed; list is a Dependency
FN-007	Referral status displayed clearly to operators	[TB section 4]	Must	Contractual; vocabulary Pending Confirmation (RT-006)
FN-008	System acknowledgement and human accept/reject recorded and displayed as separate events	D-016; RT-005	Must	Confirmed
FN-009	OpenCHS sends follow-up updates against the linked CPIMS+ case	D-012; [M1 section 7]	Must	Confirmed; update contract Pending Confirmation (RT-007)
FN-010	CPIMS+ status and intervention updates for referred cases are shown on the linked OpenCHS case	D-007; RT-008	Must	Confirmed; mechanism Pending Confirmation
FN-011	Helpline case plan is never transmitted	D-012; WF-008	Must	Confirmed
FN-012	Validation messages explain missing or invalid data	[TB section 4]	Must	Contractual

J.3 Data and field requirements

ID	Requirement	Source	Priority (proposed)	Status
DATA-001	ABR issuing authority: NICR, Ministry of Home Affairs	D-014	Must	Confirmed
DATA-002	ABR optional; receiving caseworker obtains it later if missing	D-005	Must	Confirmed
DATA-003	Remaining ABR verification contract: examples, checksum or uniqueness rules if any, invalid-value policy, live verification route	A-007	Should	Pending Confirmation

ID	Requirement	Source	Priority (proposed)	Status
DATA-004	Subsequent ABR correction and cross-system update	D-005	Should	Pending Confirmation
DATA-005	Configured child identity mapping and minimum fields	D-011; D-012	Must	Pending Confirmation (Dependency on configured definitions)
DATA-006	Age, sex, nationality, location and contact mappings	D-012	Must	Pending Confirmation
DATA-007	Case category mapped to CPIMS+ multi-select protection concerns using one standardised list	[M2 section 11]	Must	Pending Confirmation (Primero to supply list)
DATA-008	Reporter contact when available; anonymous representation	D-010; D-012	Must	Pending Confirmation (mapping)
DATA-009	Perpetrator details mapping	D-004; D-012	Must	Pending Confirmation (mapping)
DATA-010	Attachments: types, size, scanning, permissions, retention	D-012	Must	Pending Confirmation
DATA-011	Narrative mandatory at referral	D-013	Must	Confirmed
DATA-012 *(new)*	Consent or other basis for disclosure represented in CPIMS+ referral data	Primero referral consent check (Annex B.6); R-011	Must	Pending Confirmation
DATA-013 *(new)*	Stable external case reference linking the OpenCHS case and CPIMS+ case	OPS-003; [TB section 5E]	Must	Proposed

The confirmed information set and mapping status are detailed in **Annex C**.

J.4 Integration and interoperability requirements

ID	Requirement	Source	Priority (proposed)	Status
API-001	Supplied Primero API specification (API v2) is the build basis unless it changes	D-017	Must	Confirmed
API-009	Direct connection (API-to-API or OpenFN); no X-Road between OpenCHS and CPIMS+	D-015	Must	Confirmed; API vs OpenFN choice Pending Confirmation (Ministry)
GOV-004	Deployed Lesotho Primero release and base URLs recorded	—	Must	Pending Confirmation
API-002	Authentication contract for the unattended integration account	[TR1 section 3]; [TB section 5B]	Must	Pending Confirmation
API-003	Agreed operation sequence: create case, refer, follow-up update, attachments	Annex B	Must	Pending Confirmation
API-004	Error classes, timeouts, concurrency and performance expectations	[TR1 section 8]	Must	Pending Confirmation

ID	Requirement	Source	Priority (proposed)	Status
RT-007	Return-update contract: events, fields, mechanism, ordering and reconciliation	D-007	Must	Pending Confirmation
OPS-003	Idempotency key and duplicate reconciliation agreed with Primero	[TB section 5E]	Must	Pending Confirmation; candidate Proposed

J.5 Security and privacy requirements

ID	Requirement	Source	Priority (proposed)	Status
SEC-001	Lawful basis and authorised purposes for sharing confirmed	A-017; A-022	Must	Dependency
SEC-002	Reporter anonymity preserved through transfer and return updates	D-010	Must	Confirmed
SEC-003	Integration account limited to the approved mapping (least privilege)	Annex B.2	Must	Proposed
SEC-004	Credentials held in a secret store, separate per environment, with rotation	[TB section 5B]; API-008	Must	Pending Confirmation
SEC-005	Encrypted transport (proposed minimum HTTPS, TLS 1.2+, certificate verification outside local environments) and the network controls Primero specifies	[DP] Item 30; [TR1 section 3]	Must	Proposed; Primero network controls Pending Confirmation
SEC-006	Logs exclude tokens and full child payloads by default; redaction and access rules approved	[TB section 5F]; OPS-005	Must	Pending Confirmation
SEC-007	Approved synthetic child-journey dataset for testing; no real child records in test environments without authorisation	[TR1 section 9]; UAT-003	Must	Proposed
SEC-008	Return updates limited to referred cases and approved fields	D-007; RT-008	Must	Proposed (bound); direction Confirmed

J.6 Operational and reliability requirements

ID	Requirement	Source	Priority (proposed)	Status
OPS-001	Durable queue so referrals are not lost during outages; behaviour during outages agreed	[TB section 5D]; A-021	Must	Pending Confirmation
OPS-002	Retry and back-off policy with escalation after exhaustion	[TB section 5D]; delivery plan	Must	Proposed
OPS-006	Named monitoring and support owners	—	Must	Pending Confirmation
OPS-007	Incident and outage procedure	A-021; A-017	Must	Pending Confirmation
OPS-008 *(new)*	Regression testing shows existing OpenCHS functionality is not adversely affected	[TB section 6]	Must	Contractual

J.7 Reporting, audit and monitoring requirements

ID	Requirement	Source	Priority (proposed)	Status
OPS-004	Audit record per transaction: case/reference ID, ABR ID where available, transaction ID, timestamp, actor or system, endpoint/action, outcome, retry history; modelled on CPIMS+ log structure	[TB section 5F]; [M2 section 9]	Must	Confirmed in principle; final fields Pending Confirmation
AUD-001 *(new)*	Queue monitoring with controlled manual retry	[TB section 5D]	Must	Contractual
AUD-002 *(new)*	Operator-visible failure state and administrator alert on exhausted retries	Delivery plan Item 30	Should	Proposed
AUD-003 *(new)*	Helpline can review returned status and interventions per referred case	[M3 section 11]	Must	Confirmed purpose; display design Proposed

J.8 UAT and acceptance requirements

ID	Requirement	Source	Priority (proposed)	Status
UAT-001	Ministry leads UAT; participant roles as recorded	D-009	Must	Confirmed
UAT-002	Named testers, accounts and authorised signatories	—	Must	Pending Confirmation
UAT-003	Safe test data and environment approved	[TR1 section 9]	Must	Pending Confirmation
UAT-004	Core scenarios recorded and converted to numbered test cases	S2 UAT scope; [M3 section 10]	Must	Confirmed (scenarios); test cases Proposed (Annex H)
UAT-007	Defect severity and conditional go-live rules	—	Must	Pending Confirmation
UAT-010 *(new)*	Checklist positions: Child Helpline supervisor participates; BITZ provides test data and scenarios; MGYSD sign-off proposed in earlier checklist; UNICEF sign-off specified in [CT1 section 3.3]	Annotated checklist; [CT1]	Must	Reconcile Ministry endorsement; name UNICEF signatory

Annex K — Detailed connection design and security specifications

These specifications are retained from v0.8 for specialist review. They remain proposed or pending as indicated; moving them here does not establish approval or completed implementation.

K.1 Detailed connection design

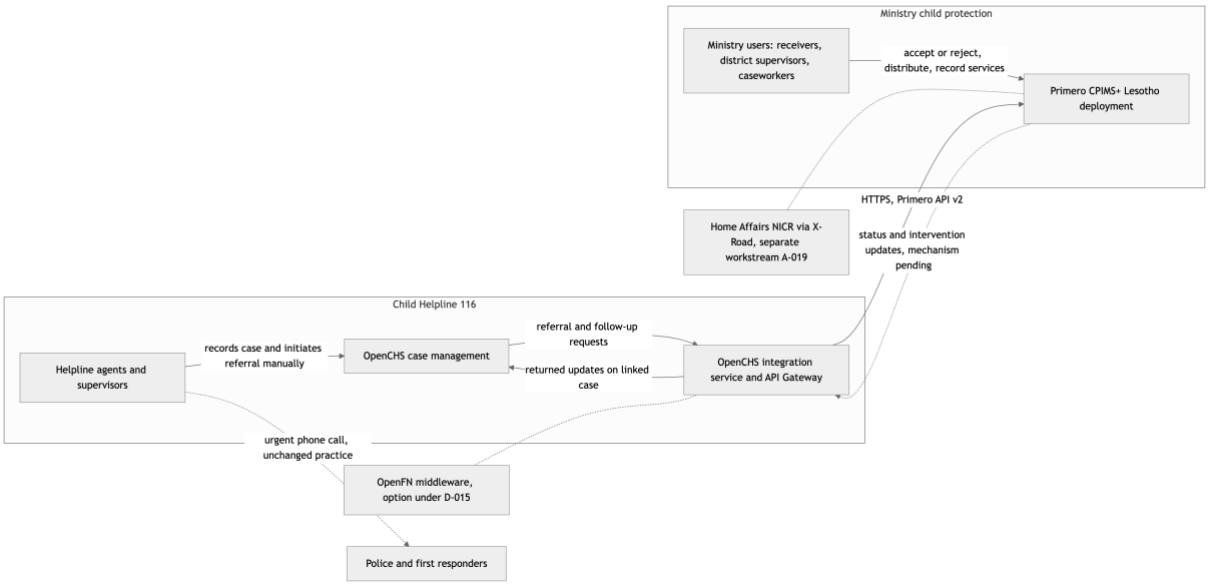
All diagrams in this section are Proposed Architecture – Subject to Technical Validation. They show Primero product capabilities documented upstream, not the deployed Lesotho configuration. OpenFN, webhooks and any X-Road arrangement are shown as options or separate workstreams, not as confirmed implementations.

K.1.1 Design principles

1. **Manual, explicit referral** — nothing is sent because a case was saved (D-006).
2. **Separate events** — system receipt, human acceptance/rejection, distribution and case progress are distinct states (D-016).
3. **No duplicates** — every retry reuses the same identifier (OPS-003).
4. **No silent loss** — failed transmissions are queued, visible and recoverable (OPS-001).
5. **Minimum necessary data** — only the agreed fields move in either direction (SEC-003, SEC-008).
6. **Evidence by design** — every transmission and update is auditable without logging sensitive content (OPS-004, SEC-006).

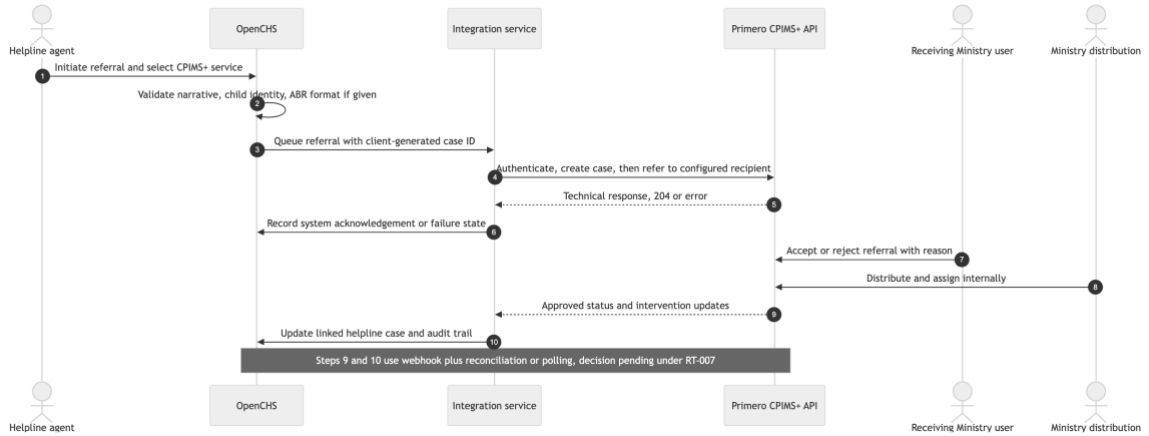
K.1.2 Diagram A — System context

Proposed Architecture – Subject to Technical Validation



K.1.3 Diagram B — Referral sequence

Proposed Architecture – Subject to Technical Validation. Operation names are from the upstream API v2 documentation (Annex B); the Lesotho operation sequence is Pending Confirmation (API-003).



K.1.4 Diagram C — Technical integration

Proposed Architecture – Subject to Technical Validation



K.1.5 Component summary

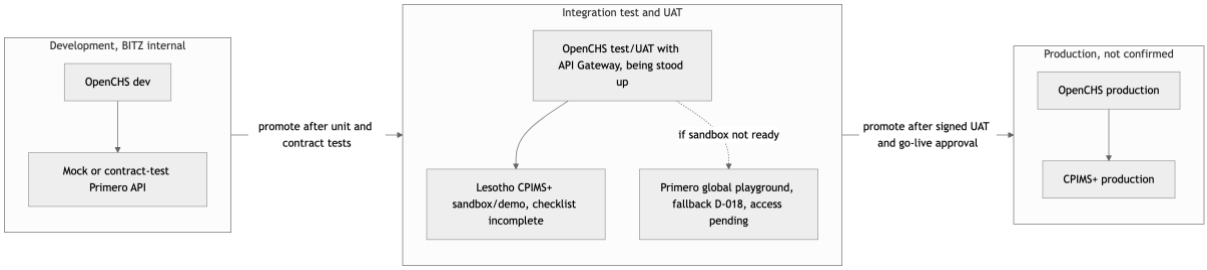
Layer	Proposed component	Evidence status
Intake	OpenCHS case form with ABR field and referral control	D2 build; form engine to verify
Outbound	Integration service via the API Gateway with durable queue, retry and idempotency	D3 build; framework and queue to verify
Authentication	Primero JWT bearer tokens; integration account	Upstream documented; Lesotho flow Pending Confirmation
Receiving system	Primero CPIMS+ API v2: case create, update, referral, attachments	Upstream documented; configuration Pending Confirmation

Layer	Proposed component	Evidence status
Return channel	Webhook with reconciliation, or polling	Options; decision Pending Confirmation (RT-007)
Controls	Audit log, monitoring, operator alerts	Design Proposed

Before design sign-off, the BITZ developers must verify the actual OpenCHS case model, integration framework, queue, role permissions, audit facilities, deployment and secret management. The Primero technical coordinator must confirm receiving-system capabilities, service configuration and permissions. Earlier technical reference material describes referral being created automatically on case save; that conflicts with D-006 and must not be implemented.

K.1.6 Diagram D — Environment architecture

Proposed Architecture – Subject to Technical Validation



Environment	Owner	Status (27 Oct 2026)	Notes
OpenCHS development	BITZ	Internal; not shared	Mock and contract tests only; must not be described as live integration
OpenCHS test/UAT with API Gateway	BITZ	Being stood up; access evidence pending (API-005)	Onboards and authenticates Primero as a connected system
Lesotho CPIMS+ sandbox/demo	Primero (Isaiah), Lesotho checklist (Rantsala)	Checklist incomplete; no progress since Session 2 (A-015)	Configured from country SOPs, forms and use cases
Primero global playground	Primero (Isaiah)	Agreed fallback (D-018); URL and access pending (A-016)	Not Lesotho-configured; tests needing country configuration must be repeated
Production (both systems)	Infrastructure owners	Not confirmed (API-008)	Endpoints, credentials, certificates and rollback to define

K.1.7 Integration reliability and audit requirements

The supplied Primero API specification is the build basis. The deployed release, authentication, operation sequence, service-to-recipient mapping and return-update mechanism require technical confirmation. The design must address durable queuing, duplicate prevention, retries, escalation, monitoring and transaction audit without asserting that those controls have been implemented. Detailed requirements are retained in Annex J.4, J.6 and J.7; API evidence and configuration dependencies are in Annex B.

K.2 Detailed data safeguards

K.2.1 Data requirements and mapping approach

The agreed referral information set includes child details, case category and narrative, reporter contact where available, perpetrator details, attachments and follow-up updates. The ABR ID remains optional and the helpline case plan is excluded. Configured identifiers, minimum fields, code lists, attachment controls and the

representation of consent remain pending. Annex C retains the mapping baseline and ABR rules; Annex J.3 retains detailed data requirements.

The received September 2019 SOP supplies Form 4 and confidentiality guidance for review. It does not establish current legal authority or the deployed CPIMS+ configuration. Ministry confirmation of applicability and approved disclosure/exception rules remains required. Section 3.5 records the evidence and its implications.

K.2.2 Legal and policy position

The sources differ on the applicable law. The supplied contract requires compliance and confirmation of data-sharing agreements but does not resolve which legal instrument is in force. In Session 3, UNICEF noted that such legislation is under development; the Session 3 action register refers to the Data Protection Act 2011; the annotated checklist refers to the Lesotho Data Protection Act 2012. **Determining the applicable legislation and its status is outside the scope of this report.** UNICEF will supply earlier Ministry data-protection work, the data management policy and legal material (A-017); BITZ will complete desk research (A-022). The responsible Ministry and UNICEF advisers must confirm the instrument, year and commencement status.

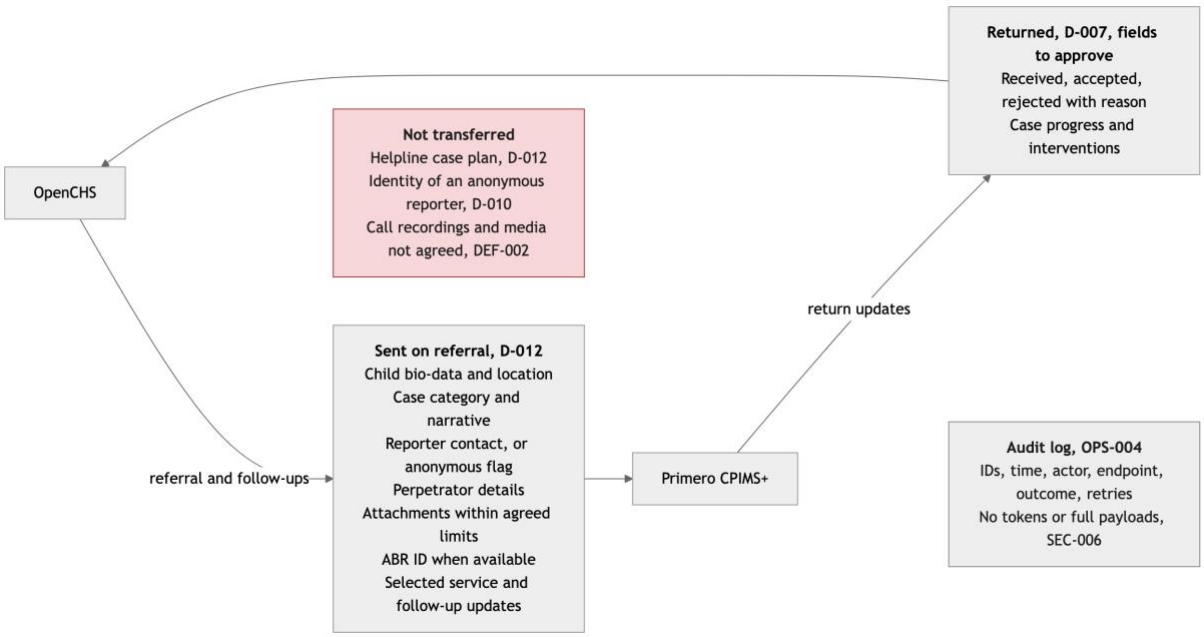
K.2.3 Control framework

Control area	Requirement	Status	Owner
Lawful basis	Authorised purposes for sharing child-protection data between the helpline and CPIMS+	Dependency (A-017, A-022)	Ministry / UNICEF
Consent and disclosure	Upstream Primero checks consent for disclosure and services at referral unless overridden; no consent capture was discussed in the sessions. Decide how consent or another basis is captured and who may override it	Pending Confirmation (R-011, DATA-012)	Ministry; Primero to confirm configuration
Anonymous reporters	Reporter recorded as anonymous in CPIMS+; anonymity preserved in payloads, logs and return updates	Confirmed (D-010); mapping Pending	BITZ, Primero
Child-sensitive data	Only agreed fields sent; no fictitious names, dates or identifiers created to satisfy validation	Confirmed field set (D-012); rule Proposed	BITZ
Access control	Integration account permitted to set only approved fields; OpenCHS role controls for referral and returned data	Proposed (SEC-003)	Primero; BITZ
Authentication	Bearer JWT, 60-minute expiry; SSO preferred for users, username/password alternative; unattended backend flow undecided	Pending Confirmation (API-002)	Primero; BITZ
Secure transport	Proposed minimum HTTPS with TLS 1.2+ and certificate verification; Primero confirmed that network/security controls apply without specifying them	Proposed; controls Pending Confirmation	Primero; infrastructure
Credential management	Secret store; separate test and production credentials; rotation; custody	Pending Confirmation (API-008)	BITZ; Primero
Audit logs	Transaction metadata logged; tokens and full payloads not logged by default; retention and access approved	Pending Confirmation (OPS-004, OPS-005)	Security owner
Data minimisation	Case plan excluded; return updates limited to referred cases and approved fields	Confirmed (D-012); bound Proposed (SEC-008)	Ministry; BITZ
Retention	Retention of queued payloads, logs and attachments	Pending Confirmation	Data protection owner

Control area	Requirement	Status	Owner
Incident handling	Contacts, escalation and notification duties for data incidents and outages	Pending Confirmation (OPS-007)	Ministry; UNICEF
Secure test data	Approved synthetic child-journey dataset; no real child records copied into test environments	Proposed (SEC-007); approval Pending	UAT lead; privacy owner
Attachments	File types, size limits, malware scanning, field permissions, retention; no call recordings unless approved	Pending Confirmation (DATA-010)	Ministry; Primero; BITZ
Return updates	Authenticated channel; minimised fields; webhook role limits fields where used	Pending Confirmation (RT-007)	Primero; BITZ

K.2.4 Data flow and classification

Technical reference — Data flow and classification. Sources: D-007, D-010, D-012, OPS-004. Which fields are returned still needs approval (RT-007).



Data	Direction	Sensitivity (proposed)	Control
Child identity, location, contact	OpenCHS → CPIMS+	High (child-identifying)	Agreed fields only; TLS; least-privilege account
Narrative, case category, perpetrator details	OpenCHS → CPIMS+	High (child-protection detail)	Not written to logs; role-based access
Reporter contact	OpenCHS → CPIMS+ when given	High	Anonymous flag respected (D-010)
Attachments	OpenCHS → CPIMS+	High	Type, size and scanning rules (DATA-010)
ABR ID	OpenCHS → CPIMS+ when available	High (national identifier)	Structural validation; stored as text
Referral status and interventions	CPIMS+ → OpenCHS	High	Approved fields only; authenticated channel

Data	Direction	Sensitivity (proposed)	Control
Audit metadata	Both systems	Medium	Retention and access approval (OPS-004, OPS-005)
Helpline case plan	Not transferred	—	Excluded from payload (D-012)

K.2.5 Matters requiring UNICEF or Ministry confirmation

1. Applicable data protection instrument and its status.
2. Lawful basis and authorised purposes for each direction of exchange.
3. Consent capture and override authority.
4. Retention periods for integration data, logs and attachments.
5. Incident notification obligations and contacts.
6. Approval of the synthetic test dataset and prohibition of real records in test.
7. Named data protection/security approver.

K.2.6 Safeguarding requirements

Reporter anonymity must be preserved in both directions. Integration permissions, transport security, secret management, log redaction, synthetic test data and restrictions on returned fields require agreed controls. Annex J.5 and Annex A.5 retain the detailed security requirements and their confirmation status.